All India Aakash Test Series for NEET - 2028

TEST - I (Code-C)

Click here for Code-D Sol.

Test Date : 30/08/2026

ANSWERS

1. (3)	37. (2)	73. (1)	109. (2)	145. (2)
2. (2)	38. (3)	74. (2)	110. (3)	146. (2)
3. (4)	39. (1)	75. (4)	111. (2)	147. (3)
4. (1)	40. (4)	76. (4)	112. (2)	148. (2)
5. (3)	41. (4)	77. (2)	113. (3)	149. (2)
6. (2)	42. (3)	78. (3)	114. (4)	150. (4)
7. (4)	43. (1)	79. (2)	115. (4)	151. (1)
8. (3)	44. (1)	80. (4)	116. (4)	152. (3)
9. (3)	45. (3)	81. (3)	117. (2)	153. (3)
10. (3)	46. (2)	82. (1)	118. (3)	154. (1)
11. (1)	47. (4)	83. (3)	119. (2)	155. (3)
12. (3)	48. (1)	84. (4)	120. (1)	156. (2)
13. (3)	49. (2)	85. (1)	121. (2)	157. (3)
14. (3)	50. (2)	86. (4)	122. (2)	158. (1)
15. (3)	51. (3)	87. (2)	123. (3)	159. (1)
16. (2)	52. (4)	88. (1)	124. (3)	160. (2)
17. (4)	53. (3)	89. (1)	125. (3)	161. (2)
18. (4)	54. (3)	90. (2)	126. (3)	162. (3)
19. (2)	55. (2)	91. (3)	127. (3)	163. (2)
20. (2)	56. (1)	92. (2)	128. (4)	164. (3)
21. (2)	57. (1)	93. (4)	129. (3)	165. (3)
22. (1)	58. (2)	94. (4)	130. (1)	166. (1)
23. (4)	59. (4)	95. (4)	131. (2)	167. (2)
24. (1)	60. (1)	96. (3)	132. (3)	168. (2)
25. (4)	61. (1)	97. (2)	133. (2)	169. (4)
26. (3)	62. (1)	98. (1)	134. (3)	170. (1)
27. (1)	63. (2)	99. (3)	135. (4)	171. (4)
28. (4)	64. (1)	100. (4)	136. (2)	172. (2)
29. (1)	65. (4)	101. (1)	137. (1)	173. (2)
30. (3)	66. (1)	102. (2)	138. (4)	174. (2)
31. (3)	67. (2)	103. (1)	139. (3)	175. (4)
32. (4)	68. (1)	104. (4)	140. (1)	176. (3)
33. (1)	69. (1)	105. (4)	141. (4)	177. (2)
34. (4)	70. (1)	106. (3)	142. (3)	178. (3)
35. (4)	71. (2)	107. (3)	143. (3)	179. (3)
36. (1)	72. (1)	108. (4)	144. (2)	180. (4)

HINTS & SOLUTIONS

[PHYSICS]

1. Answer (3)

Hint: If a body is dropped from a certain height then, time taken to reach ground, $t = \sqrt{\frac{2h}{g}}$.

Sol.: $\frac{t_1}{t_2} = \sqrt{\frac{h_1}{h_2}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$

2. Answer (2)

Hint: Least count of screw gauge

$= \frac{\text{Pitch}}{\text{Number of circular scale division}}$

Sol.: Least count $= \frac{0.75}{15} = 0.05$ mm

Area of sheet $A = Lb$

$\frac{\Delta A}{A} = \frac{\Delta L}{L} + \frac{\Delta b}{b}$

$\frac{\Delta A}{A} = \frac{0.05}{5} + \frac{0.05}{2.5}$

$\frac{\Delta A}{A} = 0.01 + 0.02$

$\frac{\Delta A}{A} = 0.03$

3. Answer (4)

Hint: Displacement $S = ut - \frac{g}{2}t^2$

Sol.: If the balls meet at time t

Then first ball will take t seconds while second ball will take $(t-2)$ seconds and they will meet at same height $\rightarrow h_1 = h_2$

$40t - \frac{g}{2}t^2 = 40(t-2) - \frac{g}{2}(t-2)^2$

$40t - \frac{g}{2}t^2 = 40t - 80 - \frac{g}{2}\left(t^2 + 4 - 4t\right)$

$40t - \frac{g}{2}t^2 = 40t - 80 - \frac{g}{2}t^2 - 2g + 2gt$

$0 = -80 - 2 \times 10 + 2 \times 10 \times t$

$\frac{100}{20} = t$

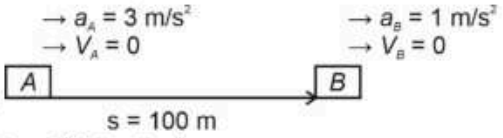
$t = 5$ s

4. Answer (1)

Hint: Relative displacement

$s_{A/B} = u_{A/B}t + \frac{a_{A/B}}{2}t^2$

Sol.:



Relative initial velocity:

$u_{A/B} = u_A - u_B = 0$

Relative acceleration:

$a_{A/B} = a_A - a_B = 3 - 1 = 2$ m/s²

By equation of motion,

$s_{A/B} = u_{A/B}t + \frac{a_{A/B}}{2}t^2$

$100 = 0 + \frac{2}{2}t^2$

$t = 10$ s

5. Answer (3)

Hint: Velocity of ball w.r.t. elevator

$V_{b/e} = V_b - V_e$

Sol.: $V_b = V_{b/e} + V_e$

$V_b = 10 + 5$

$V_b = 15$ m/s

6. Answer (2)

Hint: Relative velocity

$V_{A/B} = V_A + V_B$ [Due to opposite direction]

Sol.: $V_{A/B} = 30 + 20$

$V_{A/B} = 50$ m/s

Time of passing $T = \frac{s_{A/B}}{V_{A/B}} = \frac{100 + 150}{50}$

$T = 5$ s

7. Answer (4)

Hint: Acceleration $A = \frac{dv}{dt} = \frac{d^2x}{dt^2}$

Sol.: $x = a\left(t^2 - 2\right) + b(t-2)^2$

Velocity $v = \frac{dx}{dt} = a(2t - 0) + 2b(t - 2)$

$v = 2at + 2bt - 4b$

Acceleration $A = \frac{dv}{dt} = 2a + 2b - 0$

$A = 2(a + b)$

8. Answer (3)

Hint: Average velocity $V_m = \frac{\text{Total displacement}}{\text{Total time}}$

Sol.: $V_m = \frac{d + 1.5d}{t_1 + t_2}$

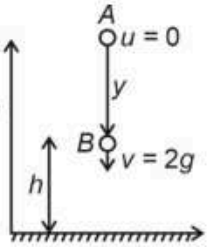
$\frac{50}{7} = \frac{2.5d}{\left(\frac{d}{5} + \frac{1.5d}{V_0}\right)}$

$V_0 = 10$ m/s

9. Answer (3)

Hint: Use $v^2 = u^2 + 2as$

Sol.:



From A to B

$v^2 = u^2 + 2as$

$(2g)^2 = 0 + 2 \times 10 \times y$

$y = 20$ m

Height from ground $h = H - y = 60 - 20 = 40$ m

10. Answer (3)

Hint: Surface Tension $T = \frac{F}{L}$

and Impulse $I = \Delta p$

Sol.: $[T] = \left[\frac{F}{L}\right] = \frac{\left[MLT^{-2}\right]}{[L]} = [MT^{-2}]$

$[I] = [\Delta p] = [MLT^{-1}]$

Dimensions of $T \neq$ dimensions of I.

11. Answer (1)

Hint: Average acceleration

$a_{avg} = \frac{V_f - V_i}{\Delta t}$

Sol.: $a_{avg} = \frac{\left(90 \times \frac{5}{18}\right) - 0}{10}$

$a_{avg} = \frac{25}{10} = 2.5\ \text{m/s}^2$

12. Answer (3)

Hint: Acceleration $\vec{a} = \frac{d\vec{v}}{dt}$

Sol.: Acceleration is due to change in velocity

- A particle can be momentarily at rest but still experience an ongoing change in velocity or acceleration.
- An object can also accelerate while moving at a constant speed due to change in direction of motion.

13. Answer (3)

Hint: Use principle of homogeneity

Sol.: $[A \sin t] = [x(t)]$

$[A]\left[M^0L^0T^0\right] = [L]$

$[A] = [L]$

Similarly

$[B] = [L]$ and $[D] = [L]$

$[Ct^2] = [x]$

$[Ct^2] = [L]$

$[C] = [LT^{-2}]$

$\therefore \left[\frac{ABC}{D}\right] = \frac{[L][L]\left[LT^{-2}\right]}{[L]}$

$\left[\frac{ABC}{D}\right] = \left[L^2T^{-2}\right] = \left[M^0L^2T^{-2}\right]$

14. Answer (3)

Hint: Relative error is given as

$\frac{\Delta Z}{Z} = \frac{\Delta p}{p} + \frac{2\Delta q}{q} + \frac{3\Delta r}{r} + \frac{1}{2}\frac{\Delta s}{s}$

Sol.: $\frac{\Delta Z}{Z} \times 100$

$= 1\% + 2 \times 2\% + 3 \times 3\% + \frac{1}{2} \times 2\%$

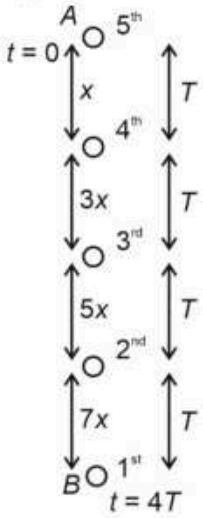
$\frac{\Delta Z}{Z} \times 100 = 1 + 4 + 9 + 1 = 15\%$

15. Answer (3)

Hint: By Galileo's ratio

$x_1 : x_2 : x_3 = 1 : 3 : 5$

Sol.:



By Galileo's ratio of odd numbers

$S_{2nd} - S_{3rd} = 5x$

$35 = 5x$

$x = 7$ m ...(i)

From A to B

$$S = ut + \frac{a}{2}t^2$$

$$16x = 0 + \frac{g}{2}(4T)^2$$

$$x = \frac{g}{2}T^2 \quad ...(ii)$$

From (i) and (ii)

$$7 = \frac{g}{2}T^2$$

$$T = \sqrt{\frac{7}{5}}\ s$$

16. Answer (2)

Hint: Use $n_1u_1 = n_2u_2$

Sol.: In one system

Force has dimensions $\rightarrow \left[M_1L_1T_1^{-2}\right]$

Dimension of force in new system $\rightarrow \left[M_2L_2T_2^{-2}\right]$

$\because \quad n_2u_2 = n_1u_1$

$$n_2 = \frac{n_1u_1}{u_2} = \frac{10\left[M_1L_1T_1^{-2}\right]}{\left[M_2L_2T_2^{-2}\right]}$$

$$n_2 = 10\left[\frac{M_1}{M_2}\right]\left[\frac{L_1}{L_2}\right]\left[\frac{T_1}{T_2}\right]^{-2}$$

$$n_2 = 10\left[\frac{1}{\alpha}\right]\left[\frac{1}{\beta}\right]\left[\frac{1}{\gamma}\right]^{-2}$$

$$n_2 = 10\alpha^{-1}\beta^{-1}\gamma^2$$

17. Answer (4)

Hint: When physical quantities are added, there should be as many decimal places retained in the final result as there are in the number with the least decimal places.

Sol.: 0.1 + 0.01 + 0.001 = 0.111 cm $\approx$ 0.1 cm

18. Answer (4)

Hint: Observed reading (R) = $MSR + n \times VC$

Sol.: $R = 2.4 + 6 \times 0.01 = 2.46$ cm

Zero correction

= –(Zero error) = –(–0.04 cm) = 0.04 cm

Correct reading (R')

R' = Observed Reading + Zero correction

$R' = 2.46 + 0.04 = 2.50$ cm

19. Answer (2)

Hint: Fractional error $\frac{\Delta g}{g} = \frac{\Delta l}{l} + \frac{2\Delta T}{T}$

Sol.: Time period of a simple pendulum

$$T = 2\pi\sqrt{\frac{l}{g}}$$

$$T^2 = 4\pi^2\frac{l}{g}$$

$$g = 4\pi^2\frac{l}{T^2}$$

$$\frac{\Delta g}{g} = \frac{\Delta l}{l} + \frac{2\Delta T}{T}$$

$$\frac{\Delta g}{g} = \frac{0.1}{40} + 2\times\frac{\Delta t}{t}$$

$$\left[\because\ t = nT, \frac{\Delta t}{t} = \frac{n\Delta T}{nT} = \frac{\Delta T}{T}\right]$$

$$\frac{\Delta g}{g}\times 100 = \left[\frac{0.1}{40} + 2\times\frac{0.5}{80}\right]\times 100 = 1.5\%$$

20. Answer (2)

Hint: Acceleration $a = \frac{dv}{dt}$

Sol.: $a = -K\sqrt{v}$

$$\frac{dv}{dt} = -K\sqrt{v}$$

$$\int_u^0 \frac{dv}{\sqrt{v}} = -\int_0^T K dt$$

$$\left[2\sqrt{v}\right]_u^0 = -KT$$

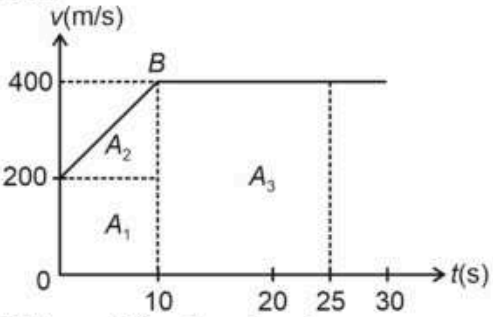
$$-2\sqrt{u} = -KT$$

$$T = \frac{2\sqrt{u}}{K}$$

21. Answer (2)

Hint: Displacement = Area under ($v - t$) graph

Sol.:



Distance (S) = $A_1 + A_2 + A_3$

$$S = 200\times10 + \frac{1}{2}\times10\times200 + (25-10)\times400$$

S = 2000 + 1000 + 6000 = 9000 m

$\therefore$ S = 9 km

22. Answer (1)

Hint & Sol.: Dimensions of change in energy will be equal to that of energy.

[Energy] = $[ML^2T^{-2}]$

23. Answer (4)

Hint: Least count of screw gauge,

$$LC = \frac{\text{Pitch}}{\text{No. of circular scale division}}$$

Sol.: Initial least count

$$LC_1 = \frac{\text{Pitch } (P)}{\text{No. of circular scale division } (N)}$$

$$LC_1 = \frac{P}{N}$$

Final least count $LC_2 = \dfrac{P + \frac{75}{100}P}{N - \frac{50}{100}N}$

$$LC_2 = \frac{\frac{7}{4}P}{\frac{N}{2}} = \frac{7}{2}\left[\frac{P}{N}\right]$$

$$LC_2 = \frac{7}{2}LC_1$$

$$LC_2 = \frac{7}{2}\times0.01 \text{ mm}$$

$$LC_2 = 35\times10^{-3} \text{ mm}$$

24. Answer (1)

Hint: Use $\Delta v = \int a\,dt$ and $\Delta x = \int v\,dt$

Sol.: Acceleration is given

$$a = \frac{dv}{dt} = 6t + 2$$

$$\int_0^v dv = \int a\,dt = \int_0^t 6t\,dt + \int_0^t 2\,dt$$

$$v = 6\left[\frac{t^2}{2}\right] + 2t$$

$$\frac{dx}{dt} = v = 3t^2 + 2t$$

$$\int_{x_i}^{x_f} dx = \int v\,dt = \int_0^2 3t^2\,dt + \int_0^2 2t\,dt$$

$$x_f - x_i = 3\left[\frac{t^3}{3}\right]_0^2 + 2\left[\frac{t^2}{2}\right]_0^2$$

Δx = 8 + 4

Δx = 12 m

25. Answer (4)

Hint: Average speed = $\dfrac{\text{Total distance}}{\text{Total time}}$

Sol.: $v_{avg} = \dfrac{2d}{\frac{d}{u} + \frac{d}{3u}} = \dfrac{2}{\frac{1}{u}\left[1 + \frac{1}{3}\right]} = \dfrac{3u}{2}$

26. Answer (3)

Hint & Sol.: $y = 3t^2 + 2t$

$$\frac{dy}{dt} = 6t + 2 = 8$$

27. Answer (1)

Hint: Volume $V = \pi r^2 L$

$$\frac{\Delta V}{V} = \frac{2\Delta r}{r} + \frac{\Delta L}{L}$$

Sol.: $\dfrac{\Delta V}{V}\times100 = \dfrac{2\times0.01}{0.5}\times100 + \dfrac{0.05\times100}{10}$

$$\frac{\Delta V}{V}\times100 = 4\% + 0.5\% = 4.5\%$$

Density of wire $d = \frac{m}{V}$

$\frac{\Delta d}{d} = \frac{\Delta m}{m} + \frac{\Delta V}{V}$

$\frac{\Delta d}{d} \times 100 = \frac{0.003}{0.6} \times 100 + 4.5\%$

$\therefore \frac{\Delta d}{d} \times 100 = 0.5\% + 4.5\% = 5\%$

28. Answer (4)

Hint: Displacement $\Delta x = \int v dt$

Sol.: Velocity $v = \alpha t - \beta t^2 = 0$

t = 0 or $t = \frac{\alpha}{\beta}$

$\frac{dx}{dt} = v = \alpha t - \beta t^2$

$\int_{x_i}^{x_f} dx = \int_0^{\alpha/\beta} \alpha t dt - \int_0^{\alpha/\beta} \beta t^2 dt$

$x_f - x_i = \alpha\left[\frac{t^2}{2}\right]_0^{\alpha/\beta} - \frac{\beta}{3}\left[t^3\right]_0^{\alpha/\beta}$

(Displacement) $d = \Delta x = \frac{\alpha^3}{6\beta^2}$

29. Answer (1)

Hint: Acceleration $a = v\left(\frac{dv}{ds}\right)$

Sol.: $v^2 = s^2 + 2s + 4$

$\frac{dv^2}{ds} = \frac{d}{ds}\left[s^2 + 2s + 4\right]$

$\frac{2vdv}{ds} = \left[2s + 2\right]$

$2a = 2s + 2$

$a = s + 1$

$a = 3 + 1$

$a = 4\ m/s^2$

30. Answer (3)

Hint: Use of equation of motion

Sol.: Acceleration $a = -g$ = constant

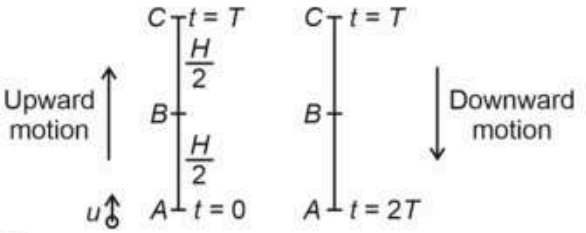
- Velocity $v = u - gt$
- Distance $S_t = u - \frac{g}{2}(2t - 1)$

It indicates that distance travelled in equal time interval decreases.

31. Answer (3)

Hint: Use symmetry between upward and downward motion

Sol.:



By symmetry

Time taken in upward motion is same as in downward motion between two given points

Time taken from A to B = Time from B to A = T_1

Time taken from B to C = Time from C to B = T_2

From $C \rightarrow A$: time taken = T

$H = ut + \frac{a}{2}t^2$

$H = 0 + \frac{g}{2}t^2$

$T = t = \sqrt{\frac{2H}{g}} = T_2 + T_1$

From $C \rightarrow B$:

$t_{C \rightarrow B} = T_2 = \sqrt{\frac{2\left(\frac{H}{2}\right)}{g}} = \frac{T}{\sqrt{2}}$

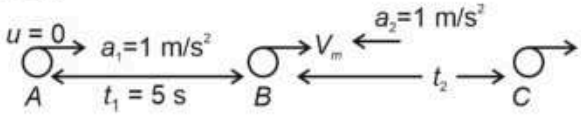
From $B \rightarrow A$:

$t_{B \rightarrow A} = T_1 = T - \frac{T}{\sqrt{2}} = \frac{\left(\sqrt{2} - 1\right)T}{\sqrt{2}} = \left(\sqrt{2} - 1\right)T_2$

32. Answer (4)

Hint: Use equations of motion

Sol.:



For $A \rightarrow B$:

$V_m = u + at_1$

$V_m = 0 + 1 \times 5$

$V_m = 5$ m/s

For $B \rightarrow C$:

$V = u + at$

$0 = 5 - 1 \times t_2$

$t_2 = 5$ s

For $A \rightarrow C$:

$S_{AC} = S_{AB} + S_{BC}$

$S_{AC} = 0 + \frac{a_1 t_1^2}{2} + V_m t_2 - \frac{a_2 t_2^2}{2}$

$S_{AC} = 0 + \frac{1}{2}(5)^2 + 5 \times (5) - \frac{1}{2}(5)^2$

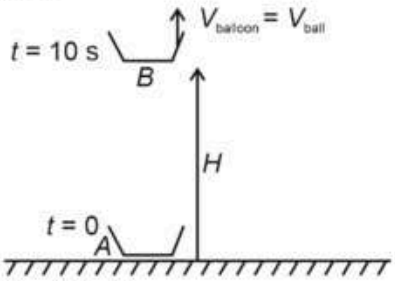
S_{AC} = 25 m

33. Answer (1)

Hint: When ball is dropped from balloon,

$V_{balloon} = V_{ball}$

Sol.:



For A to B :

$V_{Balloon} = 0 + at$

$= 2 \times 10$

$= 20$ m/s

By relative velocity

$V_{Balloon} = V_{ball} = 20$ m/s

Further height risen by ball,

$h_{max} = \frac{V_{ball}^2}{2g} = \frac{(20)^2}{2 \times 10} = 20$ m

Total height from ground, $H' = h_{max} + H$

$H' = 20 + H$ ...(i)

For balloon :

$H = 0 + \frac{a}{2} t^2$

$H = \frac{2}{2}(10)^2 = 100$ m

From (i),

$H' = 20 + 100 = 120$ m

34. Answer (4)

Hint: Use equation of motion $v^2 = u^2 - 2as$ for retardation a

Sol.: For first part:

$V^2 = u^2 - 2as$

$\left(\frac{V_0}{2}\right)^2 = V_0^2 - 2a(36)$

$\frac{3V_0^2}{4} = 72a$ ...(i)

For second part:

$0^2 = \left(\frac{V_0}{2}\right)^2 - 2as'$

$\frac{V_0^2}{4} = 2as'$ ...(ii)

From (i) and (ii)

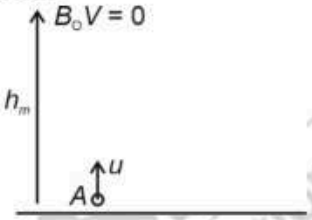
$3(2as') = 72a$

$s' = 12$ cm

35. Answer (4)

Hint: Average speed $V_{avg} = \frac{\text{Total distance}}{\text{Total time taken}}$

Sol.:



Maximum height $h_m = \frac{u^2}{2g}$

Time of flight $T = \frac{2u}{g}$

For complete journey:

Total distance $s = 2h_m$

Total time taken $T = \frac{2u}{g}$

$\therefore \quad V_{avg} = \frac{2h_m}{2u/g} = \frac{2\left(u^2/2g\right)}{2(u/g)}$

$V_{avg} = \frac{u}{2} = \frac{20}{2} = 10$ m/s

Average acceleration $a_{avg} = \frac{V_f - V_i}{T}$

$= \frac{u - (-u)}{\frac{2u}{g}} = g = 10$ m/s^2

36. Answer (1)

Hint: Use principle of homogeneity

Sol.: $\left[\frac{a}{V^2}\right] = [P]$

$[a] = [PV^2]$ ...(i)

$[b] = [V]$ ...(ii)

$$\left[\frac{a}{b^2}\right] = \left[\frac{PV^2}{V^2}\right] = [P] = [ML^{-1}T^{-2}]$$

$$[\text{Energy density}] = \left[\frac{U}{V}\right] = \frac{[ML^2T^{-2}]}{[L^3]} = [ML^{-1}T^{-2}]$$

37. Answer (2)

Hint: Use rules of determining number of significant figure

Sol.: According to rules of significant figures, insignificant zeros are three.

38. Answer (3)

Hint: Instantaneous acceleration is the rate of change of velocity

Sol.: For uniform acceleration, velocity is given as $V^2 = u^2 + 2aS$

Graph between $(V - S)$ will be parabola

- Both velocity and acceleration can change within a time interval.

39. Answer (1)

Hint: Slope of displacement time curve gives velocity.

Sol.: Ratio of velocity

$$= \frac{\tan 53^\circ}{\tan 37^\circ} = \frac{\frac{4}{3}}{\frac{3}{4}} = \frac{16}{9}$$

40. Answer (4)

Hint & Sol.: Some of the sources of systematic error are as follows:

(1) Instrumental error

(2) Imperfection in experimental technique or procedure

(3) Personal error

41. Answer (4)

Hint: Acceleration $a = \frac{dv}{dt}$

Sol.: $a = \frac{dv}{dt}$

$a = 0 - 3t^2$

$a = -3t^2$ is non-uniform

Negative sign indicates that car is undergoing retardation.

42. Answer (3)

Hint: Use principle of homogeneity

Sol.: $[At^2] = [v]$

$$[A] = \left[\frac{v}{t^2}\right] = \frac{[LT^{-1}]}{[T^2]} = [LT^{-3}]$$

$$\left[\frac{Bt}{C+t}\right] = [v] = [LT^{-1}]$$

$[B] = [LT^{-1}]$ and $[C] = [T]$

$\therefore [ABC] = [LT^{-3}][LT^{-1}][T]$

$[ABC] = [L^2T^{-3}] = [M^0L^2T^{-3}]$

43. Answer (1)

Hint: There are 7 fundamental physical quantities

Sol.: Electric current $\rightarrow$ Base quantity

Plane angle $\rightarrow$ Supplementary, derived quantity

Force $\rightarrow$ Derived quantity

44. Answer (1)

Hint & Sol.: Area under velocity time curve gives displacement. Slope of velocity time curve gives acceleration.

Area under acceleration time curve gives change in velocity.

45. Answer (3)

Hint: Stopping distance $S = \frac{u^2}{2a}$

Sol.: Distance travelled during reaction time:

$S_1 = V(\Delta t) = 30 \times 0.5 = 15$ m

Stopping distance under braking:

$$S_2 = \frac{u^2}{2a} = \frac{(30)^2}{2 \times 5} = 90 \text{ m}$$

Total stopping distance

$S = S_1 + S_2 = 15 + 90 = 105$ m

[CHEMISTRY]

46. Answer (2)

Hint: Number of atoms = Number of moles × atomicity × N_A

Sol.: Number of atoms in 32 g of

$$CH_4 = \frac{32}{16} \times 5 \times N_A = 10N_A \text{ atoms}$$

Number of atoms in 3.011×10^{23} molecules of

$$SO_2 = \frac{3.011 \times 10^{23}}{6.022 \times 10^{23}} \times N_A \times 3 = 1.5N_A \text{ atoms}$$

Number of atoms in 44.8 L of $CO_2(g)$ at STP

$$= \frac{44.8}{22.4} \times N_A \times 3 = 6N_A \text{ atoms}$$

Number of atoms in 17 g of

$NH_3(g) = \frac{17}{17} \times N_A \times 4 = 4N_A$ atoms

47. Answer (4)

Hint: $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

Sol.:

	$N_2(g)$	+ $3H_2(g)$	$\rightarrow$ $2NH_3(g)$
Mole $\Rightarrow$	1	4	

Here $N_2(g)$ is limiting reagent.

1 mole $N_2(g)$ produces 2 moles of $NH_3(g)$

Mass of NH_3 = 2 × 17 $\Rightarrow$ 34 g of NH_3

48. Answer (1)

Hint: Number of mmoles of HCl = $\frac{1}{2}$ × Number of mmoles of $CaCO_3$

Sol.: $CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$

Millimoles of HCl = 0.5 × 50 = 25 mmol

2 mmol of HCl reacts with 1 mmol of $CaCO_3$

25 mmol of HCl reacts with $\frac{1}{2}$ × 25 $\Rightarrow$ 12.5 mmol of $CaCO_3$

Mass of $CaCO_3 = \frac{12.5}{1000} \times 100 = 1.25$ g

49. Answer (2)

Hint: According to law of multiple proportions, two elements can combine to form more than one compound, the mass of one element that combines with a fixed mass of the other element, is in the ratio of small whole number.

Sol.:

$H_2C_2O_4$	+ H_2SO_4	$\rightarrow$ H_2O +	CO_2	+	CO
$\frac{9}{90} \Rightarrow 0.1$ mol			0.1 mol		0.1 mol

HCOOH	+ H_2SO_4	$\rightarrow$ H_2O +	CO
$\frac{4.6}{46} \Rightarrow 0.1$ mol			0.1 mol

Total mass of CO gas = 0.2 × 28 $\Rightarrow$ 5.6 g

Total mass of CO_2 gas = 0.1 × 44 = 4.4 g

Total weight of evolved gases = 5.6+4.4 = 10.0 g

- Total number of nucleons = $\frac{1.6}{16} \times N_A \times 16$

$\Rightarrow 1.6\ N_A$

- Significant figure in 200 eggs is ∞

50. Answer (2)

Hint: For minimum molar mass of enzyme, one zinc atom should be present

Sol.: $0.5 = \frac{1 \times 65}{\text{Minimum molar mass of enzyme}} \times 100$

Minimum molar mass of enzyme = 13000 g mol^{-1}

51. Answer (3)

Hint and Sol.: Abundance of B-10 $\Rightarrow$ x%

Abundance of B-11 $\Rightarrow$ (100 – x)%

$10.81 = \frac{10x + 11(100 - x)}{100}$

$\frac{10x + 1100 - 11x}{100} = 10.81$

x = 19%

Abundance of heavier isotope = 100 – 19 = 81%

52. Answer (4)

Hint: The prefix used for 10^{-18} multiple is atto

Sol.:

Multiple	Prefix
10^{-12}	pico
10^{12}	tera
10^{18}	exa

53. Answer (3)

Hint and Sol.: According to Dalton's atomic theory, atoms are neither created nor destroyed in a chemical reaction.

54. Answer (3)

Hint: % purity = $\frac{\text{Weight of pure substance}}{\text{Weight of impure sample}} \times 100$

Sol.: $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

Moles of $CO_2(g) = \frac{4.4}{44} = 0.1$ mol

1 mol CO_2 is liberated from 1 mol $CaCO_3$

0.1 mol CO_2 is liberated from 0.1 mol $CaCO_3$

Weight of $CaCO_3$ = 0.1 × 100 = 10 g

% purity = $\frac{10}{15} \times 100 = 66.67\%$

55. Answer (2)

Hint: Mole = $\frac{\text{Weight (in g)}}{\text{Atomic mass}}$

Sol.: Mole of P = $\frac{4}{40}$ = 0.1 mol

Mole of Q $= \frac{6.0\times10^{23}}{6.0\times10^{23}} = 1$ mol

Here R is limiting reagent

3 moles of R produce 1 mole of PQR_3

0.06 moles of R produce $\frac{1}{3}\times0.06 \Rightarrow 0.02$ mol

Atomic weight of $PQR_3 = \frac{9.4}{0.02} = 470$

$40 + x + 80\times3 = 470$

x = 190 u

56. Answer (1)

Hint: $3BaCl_2 + 2Na_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6NaCl$

Sol.: Here $BaCl_2$ is limiting reagent.

3 moles of $BaCl_2$ gives 1 mole of $Ba_3(PO_4)_2$

0.1 mole of $BaCl_2$ gives $\frac{1}{3}\times0.1 \Rightarrow 0.03$ mol of $Ba_3(PO_4)_2$

57. Answer (1)

Hint: Mole fraction of solute

$= \frac{\text{Number of moles of solute}}{\text{Total number of moles}}$

Molality $= \frac{\text{Number of moles of solute}}{\text{Weight of solvent (in kg)}}$

Sol.: 2 m means 2 mole of solute dissolve in 1000 gm of solvent.

Moles of solvent $= \frac{1000}{18} = 55.55$

Mole fraction of solute $= \frac{2}{2+55.55} \Rightarrow 0.035$

58. Answer (2)

Hint & Sol.:

$$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O$$

Moles = 1 mol 2 mol

Here NH_3 is limiting reagent

4 mol NH_3 produces 4 mol NO

1 mol NH_3 produces 1 mol NO = 30 g NO

4 mol NH_3 produces 6 mol H_2O

1 mole NH_3 produces $\frac{6}{4}$ mol H_2O

$\Rightarrow \frac{6}{4}\times18 \Rightarrow 27$ g H_2O

59. Answer (4)

Hint: Molarity (M) $= \frac{\text{Number of moles of solute}}{\text{Volume of solution (in L)}}$

Sol.: $M = \frac{3.01\times10^{22}\times1000}{6.02\times10^{23}\times200} = 0.25$ M

Molality (m) $= \frac{6.02\times10^{23}\times1000}{6.02\times10^{23}\times500} = 2$ m

60. Answer (1)

Hint: Molarity (M) $= \frac{\text{Number of moles of solute}}{\text{Volume of solution (in L)}}$

Sol.: Mole of $HNO_3 = 0.5$

Mass of $HNO_3 = 0.5 \times 63$ g

70 g HNO_3 is present in 100 g of solution

0.5 × 63 g HNO_3 is present in $\frac{100}{70}\times0.5\times63$

$\Rightarrow$ 45.0 g conc. HNO_3

61. Answer (1)

Hint: $\frac{\text{Mole of } O_2}{\text{Mole of } N_2} = \frac{\frac{2}{32}}{\frac{3}{28}} = \frac{2\times28}{3\times32} = 7:12$

Sol.: In 1 g of O_2 gas, number of atoms

$= \frac{1}{32}\times N_A\times2 = \frac{N_A}{16}$ atoms

In 1 g of O_3 gas, number of atoms

$= \frac{1}{48}\times N_A\times3 = \frac{N_A}{16}$ atoms

62. Answer (1)

Hint: Empirical formula represents the simplest whole number ratio of various atoms present in a compound.

Sol.:

Element	Percentage	Mole	Simplest ratio
C	60%	$\frac{60}{12} = 5$	$\frac{5}{2} = 2.5\times2 = 5$
H	12%	$\frac{12}{1} = 12$	$\frac{12}{2} = 6\times2 = 12$
N	28%	$\frac{28}{14} = 2$	$\frac{2}{2} = 1\times2 = 2$

Empirical formula = $C_5H_{12}N_2$

63. Answer (2)

Hint: $Ba(OH)_2 + 2HCl \rightarrow BaCl_2 + 2H_2O$

Sol.: Millimoles of H^+ ion = millimoles of HCl = 2.5 mmol

Millimoles of OH^- ion = 2 × millimoles of $Ba(OH)_2$ = 1.6 mmol

1.6 mmol of OH^- ion react with 1.6 mmol of H^+ ion.

So remaining mmol of H^+ ion = 2.5 – 1.6 = 0.9

$$[H^+] = \frac{0.9}{90} = 0.01\ M$$

64. Answer (1)

Hint: g eq. of metal = g eq. of oxygen

$$\frac{68}{E} = \frac{32}{8}$$

E = 17 g

$$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$$

Sol.: Mole ⇒ 0.5 5

Left ⇒ 0 2.5 1.5 2

Mole fraction of $CO_2 \Rightarrow \frac{1.5}{6} \Rightarrow 0.25$

65. Answer (4)

Hint: Weight of hydrated $BaCl_2$ sample - weight of dried sample = weight of H_2O

Sol.: Weight of hydrated $BaCl_2$ = 122 g

Loss in mass = 18 g

Mass of H_2O = 18 g

Moles of $H_2O = \frac{18}{18}$ = 1 mol

Gram molecular mass of $BaCl_2$ = 208 g/mol

Moles of anhydrous $BaCl_2 = \frac{104}{208} = \frac{1}{2} = 0.5$ mol

0.5 moles of $BaCl_2$ combines with = 1 mol of H_2O

1 mole of $BaCl_2$ combines with = $\frac{10}{0.5}$ = 2 mol of H_2O

So correct formula = $BaCl_2 \cdot 2H_2O$

66. Answer (1)

Hint: Energies of the orbitals in the same subshell decrease with increase in the atomic number (Z_{eff})

Sol.: $E_{2s}(H) > E_{2s}(Li) > E_{2s}(Na) > E_{2s}(K)$

67. Answer (2)

Hint: Energy in n^{th} orbit for one electron system

$$= -13.6 \times \frac{Z^2}{n^2}\ \text{eV/atom}$$

Sol.: Energy of 3rd excited state of He^+ ion

$$(E_4) = -13.6 \times \frac{4}{16} = -3.4\ \text{eV/atom}$$

Energy in 2nd excited state of H atom

$$(E_3) = -13.6 \times \frac{1}{9} = -1.51\ \text{eV/atom}$$

Energy in 4th Bohr orbit of Be^{3+}

$$(E_4) = -13.6 \times \frac{16}{16} = -13.6\ \text{eV/atom}$$

68. Answer (1)

Hint: The total number of orbitals associated with the principal quantum number (n) is n^2

Sol.: The degeneracy of the third excited state of H atom will be $(4)^2 \Rightarrow 16$

69. Answer (1)

Hint: In photoelectric effect, electrons were ejected when certain metals like potassium, caesium were exposed to a beam of light.

Sol.: The kinetic energy of the ejected electrons increases with increase of frequency of the light used.

70. Answer (1)

Hint: $\frac{1}{\lambda} = R_H Z^2 \left[\frac{1}{n_i^2} - \frac{1}{n_f^2}\right]$

Sol.: $\frac{1}{\lambda_1} = R_H \left[\frac{1}{9} - \frac{1}{16}\right]$

$$\frac{1}{\lambda_1} = R_H \times \frac{7}{16 \times 9} \quad ...(1)$$

$$\frac{1}{\lambda_2} = R_H \left[\frac{1}{16} - \frac{1}{25}\right]$$

$$\frac{1}{\lambda_2} = R_H \times \frac{9}{16 \times 25} \quad ...(2)$$

Dividing equation (1) by equation (2)

$$\frac{\lambda_2}{\lambda_1} = \frac{\frac{7R_H}{16 \times 9}}{\frac{9R_H}{16 \times 25}}$$

$$\frac{\lambda_2}{\lambda_1} = \frac{7 \times 16 \times 25}{16 \times 9 \times 9}$$

$$\frac{\lambda_2}{\lambda_1} = \frac{175}{81}$$

71. Answer (2)

Hint: Orbital angular momentum $= \hbar\sqrt{l(l+1)}$

Sol.: For p orbital, l = 1

Orbital angular momentum $= \hbar\sqrt{1(1+1)} = \hbar\sqrt{2}$

72. Answer (1)

Hint: Radial nodes = (n – l – 1)

Sol.:

Orbital	Radial node(s)
4*p*	2
4*d*	1
5*s*	4
4*f*	0

73. Answer (1)

Hint: de Broglie wavelength $(\lambda) = \dfrac{h}{mv}$

Sol.: $\lambda = \dfrac{2\pi r}{n}$ ...(1)

$r_4 = a_0 \times \dfrac{16}{1}$ ...(2)

Putting the value of r_4 in equation (1)

$\lambda = \dfrac{2\pi \times 16a_0}{4}$

$\lambda = 8\pi a_0$

74. Answer (2)

Hint: The value of m_l = –l to +l including zero

Sol.: For n = 4

l = 0, 1, 2, 3

75. Answer (4)

Hint: The maximum number of electrons in the shell with principal quantum number *n* is equal to $2n^2$

Sol.:

Work function (w_0)	Metal
2.42	Li
3.7	Mg

Balmer spectral line lies in visible region while Paschen spectral line lies in infrared region.

76. Answer (4)

Hint: d_{z^2} and $d_{x^2-y^2}$ orbitals have electron density along the axes.

Sol.: Two electrons occupying the same orbital are distinguished by spin quantum number.

77. Answer (2)

Hint: The energy of an electron in a hydrogen atom is determined solely by the principal quantum number.

Sol.: The correct order of energies of atomic orbitals for hydrogen atom is

$1s < 2s = 2p < 3s = 3p = 3d < 4s$

78. Answer (3)

Hint: de Broglie wavelength $(\lambda) = \dfrac{h}{mv}$

Sol.: $\lambda = \dfrac{6.626 \times 10^{-34}\ \text{J s}}{10 \times 10^{-3} \times 10^{-3} \times 100}$

$\lambda = 6.626 \times 10^{-31}\ \text{m}$

79. Answer (2)

Hint: The lower the value of (n + l) for an orbital, the lower is its energy. If two orbitals have the same value of (n + l), then the orbital with lower value of n will have the lower energy.

Sol.:

Orbital	(n + l) value
5*d*	7
5*f*	8
7*s*	7
6*p*	7

80. Answer (4)

Hint: When electromagnetic radiations are arranged in their increasing order of wavelength, then the spectrum which is obtained is called the electromagnetic spectrum.

Sol.: Correct order of wavelength of electromagnetic radiations

= IR > UV > X rays > γ rays

81. Answer (3)

Hint: $E = h\nu_0 + KE$

Sol.: $\dfrac{hc}{\lambda} = w_0 + 1.74 \times 10^5$

$w_0 = \dfrac{6.6 \times 10^{-34} \times 3 \times 10^8 \times 6 \times 10^{23}}{300 \times 10^{-9}} - 1.74 \times 10^5$

$w_0 = 2.2 \times 10^5\ \text{J}$

82. Answer (1)

Hint: Hund's rule states that "pairing of electrons in the orbitals belonging to the same subshell does not take place until each orbital belonging to that subshell has got one electron each."

Sol.: 

It does not follow Hund's rule of maximum multiplicity.

83. Answer (3)

Hint: The general electronic configuration of Cu atom is $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^6$, $4s^1$, $3d^{10}$

Sol.: 4s and 3p orbitals have (n + l) value four. So total 7 electrons are present.

84. Answer (4)

Hint: Isoelectronic species have same number of electrons.

Sol.: NO^+, CO and CN^- have 14 electrons

85. Answer (1)

Hint: $\Delta E = R_H Z^2 \left[\frac{1}{n_i^2} - \frac{1}{n_f^2}\right]$

Sol.: For $n = 5 \rightarrow 4$

$$\Delta E = R_H\left[\frac{1}{16} - \frac{1}{25}\right]$$

$$\Delta E = \frac{R_H \times 9}{16 \times 25}$$

For $n = 6 \rightarrow 4$

$$\Delta E = R_H\left[\frac{1}{16} - \frac{1}{36}\right]$$

$$\Delta E = \frac{20R_H}{16 \times 36}$$

For $n = 5 \rightarrow 3$

$$\Delta E = R_H\left[\frac{1}{9} - \frac{1}{25}\right] = \frac{16R_H}{9 \times 25}$$

For $n = 6 \rightarrow 1$

$$\Delta E = R_H\left[\frac{1}{1} - \frac{1}{36}\right] = \frac{35R_H}{36}$$

86. Answer (4)

Hint: According to Heisenberg's uncertainty principle, $\Delta x \times \Delta p \geq \frac{h}{4\pi}$

Sol.: $\Delta x = 2\Delta p$

$$\Delta p \times 2\Delta p = \frac{h}{4\pi}$$

$$2m^2\Delta v^2 = \frac{h}{4\pi}$$

$$\Delta v = \frac{1}{2m}\sqrt{\frac{h}{2\pi}}$$

$$\frac{r_3}{r_4} = \frac{a_0 \times \frac{9}{3}}{a_0 \times \frac{16}{4}}$$

$$\frac{x}{r_4} = \frac{3}{4}$$

$$r_4 = \frac{4x}{3}$$

87. Answer (2)

Hint: Total number of spectral lines $= \frac{\Delta n(\Delta n + 1)}{2}$

Sol.: 5th excited state, n = 6

1st excited state, n = 2

Total spectral lines = $\frac{4 \times 5}{2} = 10$

88. Answer (1)

Hint: For Lyman series $n_i = 1$

For Balmer series $n_i = 2$

Sol.: $\left(\frac{1}{\lambda_1}\right)_{Lyman} = R_H \times (1)^2\left[\frac{1}{1} - \frac{1}{4}\right]$

$$\left(\frac{1}{\lambda_1}\right)_{Lyman} = \frac{3R_H}{4} \quad ...(1)$$

$$\left(\frac{1}{\lambda_2}\right)_{Balmer} = R_H \times (1)^2\left[\frac{1}{4} - \frac{1}{16}\right]$$

$$\left(\frac{1}{\lambda_2}\right)_{Balmer} = \frac{12R_H}{64} \quad ...(2)$$

Equation (1) is divided by equation (2)

$$\frac{\frac{1}{\lambda_1}}{\frac{1}{\lambda_2}} = \frac{\frac{3R_H}{4}}{\frac{12R_H}{64}}$$

$$\frac{\lambda_2}{\lambda_1} = \frac{64 \times 3}{12 \times 4}$$

$$\frac{\lambda_2}{x} = 4$$

89. Answer (1)

Hint: Mass number = Total number of neutrons + Total number of protons

Sol.: Number of electrons = X

Number of neutrons = X + 0.304X

= 1.304X

Number of protons = X + 3

56 = X + 3 + 1.304X

Number of protons = 26

So correct symbol is $^{56}_{26}X^{3+}$

90. Answer (2)

Hint: Cathode rays consists of negatively charged particle called electrons.

Sol.: The characteristics of cathode rays do not depend upon the material of electrodes and the nature of the gas present in the cathode ray tube.

[BOTANY]

91. Answer (3)

Hint: Human RBCs contain approximately 52% proteins in the membrane.

Sol.:

Mitochondria	–	Double membrane bound structure that divide by fission
Mycoplasma	–	0.3 μm in length and lacks cell wall
Viruses	–	Non-cellular, exception to cell theory

92. Answer (2)

Hint: Robert Brown described the nucleus as early as in 1831.

Sol.:

Golgi apparatus	–	First observed in 1898 near the nucleus by Camillo Golgi.
Plasma membrane	–	Fluid mosaic model was proposed in 1972
Ribosomes	–	First observed under electron microscope as dense particles in 1953

93. Answer (4)

Hint: A special membranous structure is the mesosome which is formed by the extensions of plasma membrane into the cell and help in cell wall formation.

Sol.:

Fimbriae	–	Small bristle like fibres, do not play role in motility.
Glycocalyx	–	Protects the cells and also helps in adhesion.
Contractile vacuole	–	Found in *Amoeba*

94. Answer (4)

Hint: Chromatin is found in non-dividing cells nucleus.

Sol.: During different stages of cell division, cells show chromosomes in place of nucleus.

95. Answer (4)

Hint: Plastids are found in almost all plant cells and in euglenoids.

Sol.: Nucleus is present in WBCs. Cilia is absent in prokaryotes. Polysome is found in prokaryotes and synthesises proteins.

96. Answer (3)

Hint: Anthocyanin is a water soluble pigment.

Sol.: Microfilaments help in pseudopodia formation. Cisternae of Golgi apparatus resemble with SER.

97. Answer (2)

Hint: Antonie Von Leeuwenhoek first saw and described a live cell.

Sol.: Rudolf Virchow gave *Omnis cellula-e cellula i.e.* cells arise from pre-existing cells.

98. Answer (1)

Hint: Plastids are characteristics of plant cells.

Sol.: Animal cells have centrioles which are absent in almost all plant cells. Large sap vacuoles are found in plant cells.

99. Answer (3)

Hint: Acrocentric chromosome have sub-terminal centromere.

Sol.: Acrocentric chromosomes appear J-shaped during anaphase.

100. Answer (4)

Hint: Cell wall is the outermost covering of plant cell.

Sol.: Primary cell wall is formed towards outer side of plasma membrane.

101. Answer (1)

Hint: Free ribosomes synthesize non-secretory proteins.

Sol.: Plasma membrane have functions like cell growth, formation of intercellular junctions, secretion, endocytosis and cell division.

102. Answer (2)

Hint: In prokaryotes, chemically complex envelope system is found.

Sol.: Primary lysosome contain inactive hydrolytic enzymes.

In polysome, several ribosomes may attach to a single mRNA.

103. Answer (1)

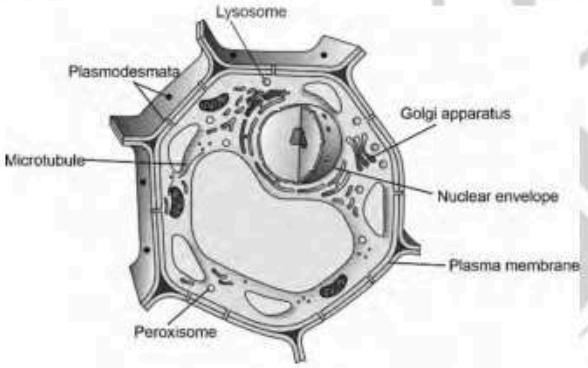
Hint: Endomembrane system include single membrane bound organelles.

Sol.: Carbohydrate synthesis is the function of chloroplast.

104. Answer (4)

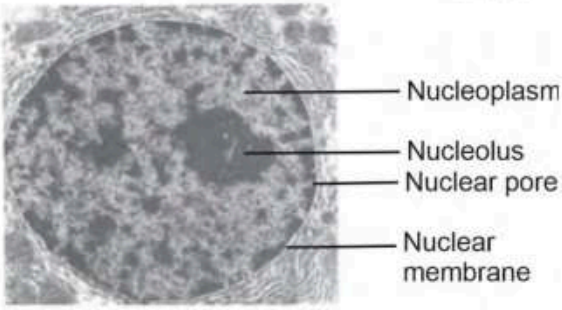
Hint: The given diagram is of plant cell because there is presence of large central vacuole.

Sol.:



105. Answer (4)

Hint:



Structure of nucleus

Sol.: Nucleolus is the site for active synthesis of rRNA. Nuclear pores allow the movement of RNA and proteins in both directions.

106. Answer (3)

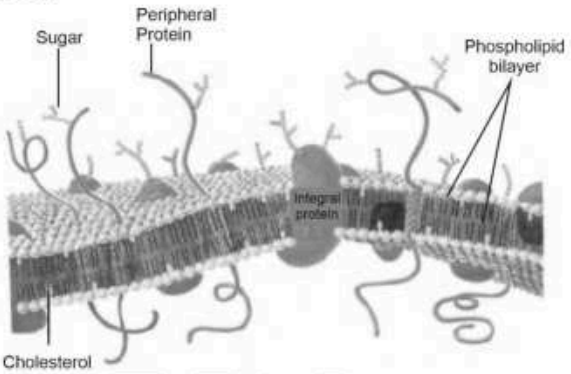
Hint: In a eukaryotic cell, ribosomes are not found in nucleus.

Sol.: Mitochondria does not possess sugar synthesising machinery.

107. Answer (3)

Hint: The given structure is of fluid mosaic model of plasma membrane, proposed by Singer and Nicolson (1972).

Sol.:



Integral protein which run throughout lipid bilayer cannot be removed easily but peripheral proteins can be removed. Lipids perform flip-flop movement.

108. Answer (4)

Sol.: The two ribosomal subunits remain united due to specific concentration of Mg^{++}.

109. Answer (2)

Hint: In centrosome, both the centrioles lie perpendicular to each other in which each has an organisation like cartwheel.

Sol.: Centrosome is non-membrane bound organelle.

110. Answer (3)

Hint: Chloroplast perform various functions like photosynthesis and storage of starch.

Sol.: Mitochondria performs oxidative phosphorylation. RER gives rise to SER.

111. Answer (2)

Hint: Svedberg's unit is indirectly a measure of density and size of ribosomal subunits.

Sol.: Onion peel cell has cell wall as the outer boundary. Prokaryotic ribosomes can be found in cytoplasm.

112. Answer (2)

Hint: Golgi apparatus is responsible for the synthesis of glycoproteins and glycolipids.

Sol.: In fungi, cell wall is made up of chitin. In algae, it is made up of cellulose, galactans mannans and calcium carbonate. In plant, it is made up of cellulose, hemicellulose, pectin and proteins.

113. Answer (3)

Hint: ER and Golgi apparatus are components of endomembrane system.

Sol.: Materials to be packaged in the form of vesicles from ER fuse with convex face of Golgi apparatus and move towards the concave face.

114. Answer (4)

Hint: Both centriole and flagella in eukaryotes have 9 radial spokes and are made up of microtubules.

Sol.: Eukaryotic flagella shows 9 + 2 arrangement of microtubules while centriole exhibits 9 + 0 arrangement of microtubules.

115. Answer (4)

Hint: In chloroplast, pigments (chlorophyll) are present in thylakoids.

Sol.: Chloroplasts can be lens-shaped, oval, spherical, discoid or even ribbon like organelles. In BGA, chloroplasts are absent.

116. Answer (4)

Hint: Lysosomes are also called suicidal bags.

Sol.: Tonoplast is membrane of vacuole. Vacuole occupy up to 90% cell volume in plants.

117. Answer (2)

Hint: Plasma membrane is selectively permeable in nature.

Sol.: Na^+/K^+ pump is an example of active transport.

118. Answer (3)

Hint: Cytoplasm is the main arena of all cellular activities.

Sol.: Plasmid DNA is used to monitor bacterial transformation.

119. Answer (2)

Hint: Reserve material in prokaryotes are stored in cytoplasm in the form of inclusion bodies.

Sol.: Polysome is not an inclusion body.

120. Answer (1)

Hint: Cristae is infolding of mitochondria, cilia is found in eukaryotic cells and cisternae in ER and Golgi complex.

Sol.: Mesosome increases the surface area for the metabolic activities occurring in prokaryotic cell.

121. Answer (2)

Hint: The elaioplasts store fats and oils.

Sol.: Amyloplasts store starch.

122. Answer (2)

Hint: Centromere appears as a narrow region called primary constriction.

Sol.: Kinetochore is the disc shaped structure present on the sides of centromere.

123. Answer (3)

Hint: Dictyosomes is not a part of chromosome.

Sol.: Telomeres seal the ends of chromosomes and prevent their shortening.

124. Answer (3)

Hint: Middle lamella is made up of calcium and magnesium pectate and hold the adjacent cells together.

Sol.: Plasmodesmata forms the symplastic system between two cells.

125. Answer (3)

Hint: Mitochondria and centrosome are not components of endomembrane system.

Sol.: RER provides the precursors of enzymes for lysosome formation in Golgi complex.

126. Answer (3)

Hint: Cisternae is structural component of ER and Golgi complex.

Sol.: Cristae or inner face of inner membrane of mitochondria is studded with oxysomes.

127. Answer (3)

Hint: This organelle also form spindle fibres that gives rise to spindle apparatus.

Sol.: The centriole is involved in basal body formation in cilia and flagella of eukaryotes.

128. Answer (4)

Hint: Ribosome is non-membrane bound organelle.

Sol.: Lysosome exhibits polymorphism and is single membrane bound.

129. Answer (3)

Hint: Some Golgian vacuole may form lysosome.

Sol.: Lysosome are formed by the process of packaging in Golgi apparatus.

130. Answer (1)

Hint: 60S larger subunit of ribosomes consists of about 40 proteins and three types of rRNAs.

Sol.: 40S smaller subunit consists with 33 protein molecules and a single 18S – rRNA.

131. Answer (2)

Hint: Chromatophores are involved in photosynthesis.

Sol.: Mesosome and mitochondria play important function in respiration.

132. Answer (3)

Hint: Nucleus is known as brain of the cell.

Sol.: Mitochondria is the powerhouse of the cell.

133. Answer (2)

Hint: The organelles whose functions are coordinated form endomembrane system.

Sol.: Centrosome is not included in endomembrane system.

134. Answer (3)

Hint: Gas vacuoles are not surrounded by unit membrane.

Sol.: Gas vacuoles are found in BGA, purple and green photosynthetic bacteria.

135. Answer (4)

Hint: Red blood cells (human) are about 7 μm in diameter.

Sol.: Egg of Ostrich is the largest isolated single cell.

[ZOOLOGY]

136. Answer (2)

Hint: Supportive cells

Sol.: Neuroglia make up more than one-half the volume of neural tissue in our body.

Chondrocytes are the cells present in cartilage.

Neurons are the structural and functional unit of the nervous system.

Fibroblasts are found in the connective tissues.

137. Answer (1)

Hint: Base + Sugar = Nucleoside

Sol.: Nitrogenous base is a nitrogen-containing ring structure that acts as the alphabet of genetics. There are two main types:

- Purines: Adenine (A) and Guanine (G).
- Pyrimidines: Cytosine (C), Thymine (T) (found in DNA) and Uracil (U) (found in RNA).
- Nucleoside is a structural subunit formed by joining of a nitrogenous base to a five-carbon pentose sugar (Ribose in RNA and Deoxyribose in DNA) *via* a glycosidic bond.
- Nucleotide : It is the foundational building block of DNA and RNA. It is formed when phosphate group is attached to the 5′ carbon of a nucleoside *via* an ester bond.

  Nucleotide = Nucleoside + Phosphate group
- Nucleic acid : A long, continuous polymer formed by linking multiple nucleotide monomers together. The phosphate group of one nucleotide connects to the 3′ carbon sugar of the next, creating a strong sugar-phosphate backbone.

138. Answer (4)

Hint: Different types of epithelium are present.

Sol.: Epithelial tissue has a free surface, which faces either a body fluid or the outside environment and thus provides a covering or a lining for some part of the body. The cells are compactly packed with little intercellular matrix. There are two types of epithelial tissues, namely, simple and compound epithelium. Simple epithelium is composed of a single layer of cells and functions as a lining for body cavities, ducts, and tubes. The compound epithelium consists of two or more cell layers and has protective function. Squamous, cuboidal and columnar epithelium are present in our body.

139. Answer (3)

Hint: Bone is a specialized connective tissue.

Sol.: Fibres and fibroblasts are compactly packed in the dense connective tissues. Orientation of fibres show a regular or irregular pattern and are called dense regular and dense irregular tissues, respectively. In the dense regular connective tissues, the collagen fibres are present in rows between many parallel bundles of fibres. Tendons, which attach skeletal muscles to bones, and ligaments which attach one bone to another are examples of this tissue. Dense irregular connective tissue has fibroblasts and many fibres (mostly collagen) that are oriented differently.

140. Answer (1)

Hint: Smooth in appearance

Sol.: The wall of internal organs like stomach, intestine, *etc.*, possess smooth muscle fibres.

The smooth muscle fibres do not show striations and possess cell junctions. Skeletal muscle fibres are multi-nucleated.

141. Answer (4)

Hint: Equal to the number of ears present in a man

Sol.: Mast cells are immune cells found in almost all connective tissues. They produce histamine, heparin and serotonin. They are not found in blood.

142. Answer (3)

Hint: The signal is transmitted through synapses between neurons.

Sol.: When a neuron is suitably stimulated, an electrical disturbance is generated which swiftly travels along its plasma membrane. Arrival of the disturbance at the neuron's endings, or output zone, triggers events that may cause stimulation or inhibition of the adjacent neurons and other cells.

143. Answer (3)

Hint: Present at locations like alveolar membrane and Bowman's capsule.

Sol.: Squamous epithelium facilitates diffusion and filtration because it consists of a single, ultra-thin layer of flat cells. This minimal thickness creates the shortest possible distance for molecules to travel, allowing for the rapid, passive exchange of gases, nutrients, and waste products.

144. Answer (2)

Hint: Proteins efficiently function at the optimum temperature.

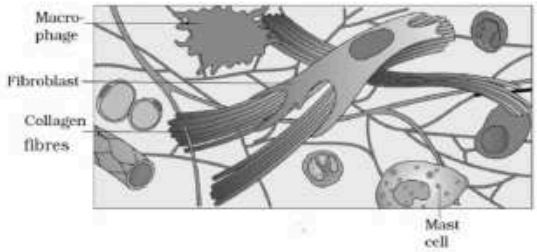
Sol.: Mostly enzymes are proteinaceous. Their function is entirely dependent on their specific three-dimensional [tertiary] shape. When exposed to extremely high temperature [195°C], the following occurs:

- Denaturation: High temperature breaks the weak hydrogen bonds and disulfide bridges that maintain the protein's complex folding.
- Once the tertiary structure is disrupted, the enzyme can no longer function properly.
- Proteinaceous enzymes contain peptide bonds. At low temperature, the enzymes get inactivated. Inactivated enzymes can regain their activity.

145. Answer (2)

Hint: Multiple of 2

Sol.: Erythrocytes (RBCs) and osteocytes (bone cells) are absent in loose connective tissues. Loose connective tissue has cells and fibres loosely arranged in a semi-fluid ground substance. They contain fibroblasts (cells that produce and secrete fibres), macrophages plasma cells and mast cells.



146. Answer (2)

Hint: Hormones are secreted by endocrine glands.

Sol.: Exocrine secretions ⇒ Mucus, saliva, oil, milk and digestive enzymes.

- Moist surface of buccal cavity possesses compound epithelium.
- Endocrine glands secrete hormones.
- Goblet cells of the alimentary canal are unicellular and isolated glandular cells.

147. Answer (3)

Hint: Cilia move particles or mucus in a specific direction.

Sol.: Compound epithelium is made of more than one layer (multi-layered) of cells and thus, has a limited role in secretion and absorption. Their main function is to provide protection against chemical and mechanical stresses. The columnar epithelium is composed of a single layer of tall and slender cells. Their nuclei are located at the base. The cuboidal epithelium is composed of a single layer of cube-like cells. This is commonly found in ducts of glands and tubular parts of nephrons in kidneys and its main functions are secretion and absorption. The epithelium of proximal convoluted tubule (PCT) of nephron in the kidney has microvilli. If the columnar or cuboidal cells bear cilia on their free surface, they are called ciliated epithelium. Their function is to move particles or mucus in a specific direction over the epithelium.

148. Answer (2)

Hint: Connect cells

Sol.: **Adhering junctions:** Their primary role is to perform cementing to keep neighbouring cells together. They provide strong mechanical attachment and structural unity to the tissue.

Tight junctions: These are responsible for preventing substances (like ions and molecules) from leaking across the epithelial layer.

Gap junctions: These facilitate communication between cells by connecting the cytoplasm of adjoining cells.

149. Answer (2)

Hint: It is a type of dense regular connective tissue.

Sol.: In the dense regular connective tissues, the collagen fibres are present in rows between many parallel bundles of fibres. Tendons, which attach skeletal muscles to bones, and ligaments which attach one bone to another are examples of dense regular connective tissues.

Fibroblasts are present in ligaments.

Bone is the site for production of RBCs.

150. Answer (4)

Hint: Has tall and slender cells

Sol.: The cuboidal epithelium is composed of a single layer of cube-like cells. This is commonly found in ducts of glands, thyroid follicles and tubular parts of nephrons in kidneys and its main functions are secretion and absorption. The columnar epithelium is composed of a single layer of tall and slender cells. They are found in the lining of stomach and intestine and help in secretion and absorption.

151. Answer (1)

Hint: Major role is protection

Sol.: Compound epithelium is made of more than one layer (multi-layered) of cells and thus, has a limited role in secretion and absorption. They cover the dry surface of the skin, pharynx, inner lining of ducts of salivary gland and of pancreatic ducts.

152. Answer (3)

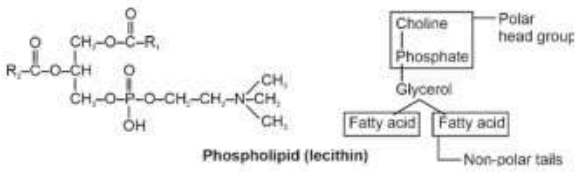
Hint: Cartilage resists compression.

Sol.: The intercellular matrix of cartilage is solid and pliable and resists compression. Cells of this tissue (chondrocytes) are enclosed in small cavities within the matrix secreted by them. Bones have a hard and non-pliable ground substance rich in calcium salts and collagen fibres which give bone its strength. It is the main tissue that provides structural frame to the body. The bone cells (osteocytes) are present in the spaces called lacunae. Bones interact with skeletal muscles attached to them to bring about movements. The bone marrow in some bones is the site of production of blood cells.

153. Answer (3)

Hint: Lecithin is an amphiphilic molecule.

Sol.:



Phospholipid (lecithin)

154. Answer (1)

Hint: Bronchioles have ciliated epithelium.

Sol.: The inner surface of the fallopian tube is lined by ciliated columnar epithelium. The function of hair-like cilia is to move the ovum (egg) through the oviduct.

The wall of the fallopian tube is composed of smooth muscle. This muscle performs involuntary peristaltic movements (rhythmic contractions) to assist in transport of the ovum.

155. Answer (3)

Hint: More than the number of eyes present in a human

Sol.: Cardiac and skeletal muscle fibres (biceps) are cylindrical in shape and have striated appearance.

- Biceps (skeletal muscles) are multinucleated, unbranched, lack gap junctions and intercalated discs.
- Cardiac muscle fibres are uninucleated, branched and have gap junctions and intercalated discs.

156. Answer (2)

Hint: In unicellular organisms, all functions are performed by a single cell.

Sol.: Cells, tissues, organs and organ systems split up the work in a way that exhibits division of labour and contributes to the survival of the body as a whole.

All multicellular animals show division of labour. In unicellular organisms, like *Paramecium*, all functions like digestion, respiration, and reproduction are performed by a single cell and thus, they do not show division of labour.

157. Answer (3)

Hint: Also found in the inner lining of fallopian tube

Sol.: Ciliated epithelium is mainly present in the inner surface of hollow organs like bronchioles and fallopian tubes. Their function is to move particles or mucus in a specific direction over the epithelium. Cuboidal epithelium is found in PCT.

158. Answer (1)

Hint: Glycogen is stored in liver.

Sol.: Glycogen is a storage polysaccharide in animals.

Maltose contains two glucose units and has a free aldehyde group hence, it acts as a reducing sugar. As both the aldehyde and ketone groups are involved in glycosidic bond formation, sucrose is a non-reducing sugar.

Cellulose cannot be digested by humans due to lack of cellulase enzyme.

159. Answer (1)

Hint: Different types of amino acids are present in proteins.

Sol.: Denaturation of proteins affects secondary and tertiary structures of proteins. Primary structure remains intact during the process of denaturation.

Each protein is a polymer of amino acids.

The amino acids in a protein can be essential or non-essential.

A peptide bond is a covalent linkage that is formed by condensation or dehydration reaction, where the carboxyl group of one amino acid reacts with the amino group of another, releasing a single molecule of water as a byproduct.

160. Answer (2)

Hint: Insulate the adjacent neurons

Sol.: Reception of stimuli, generation and transmission of nerve impulses are functions of neurons. Neurons, the units of neural system, are excitable cells. The neuroglial cells, which constitute the rest of the neural system, protect and support neurons. Neuroglial cells are non-excitable cells.

161. Answer (2)

Hint: Molecular weight of lipids is less than 800 Da.

Sol.: Lipids are not considered as polymeric substances as they do not contain long repeating chains of identical/different monomers. While they are large molecules (macromolecules) found in the acid-insoluble fraction, they are formed by smaller, distinct subunits (fatty acids + glycerol) rather than a repeating polymeric chain. Glycosidic bonds (covalent bonds) are present in carbohydrates. Lipids have ester bonds (covalent bonds).

162. Answer (3)

Hint: Rubber is a polymeric substance.

Sol.:

Pigments	Carotenoids, Anthocyanins, *etc.*
Alkaloids	Morphine, Codeine, *etc.*
Terpenoides	Monoterpenes, Diterpenes *etc.*
Essential oils	Lemon grass oil, *etc.*
Toxins	Abrin, Ricin
Lectins	Concanavalin A
Drugs	Vinblastin, curcumin, *etc.*
Polymeric substances	Rubber, gums, cellulose

163. Answer (2)

Hint: Specific pocket or cleft in the enzyme's structure.

Sol.: Enzymes are highly specific due to their unique, three-dimensional structure, particularly the shape and chemical environment of their active site. Only specific substrate molecules fit into the active site of the enzyme, similar to lock and key.

164. Answer (3)

Hint: Enzymes that catalyse the hydrolysis of peptide bond.

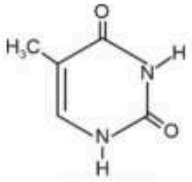
Sol.: Pepsin is a vital gastric enzyme that breaks down proteins into its smaller units. Pepsin belongs to class III (Hydrolases) of enzymes.

Hydrolases are the set of enzymes that catalyse hydrolysis of ester, ether, peptide, glycosidic, C–C, C–halide, P–N bonds, *etc.*

165. Answer (3)

Hint: Uracil is present only in RNA.

Sol.: Thymine is a substituted pyrimidine present in DNA, pairing exclusively with adenine (A) *via* two hydrogen bonds. It provides structural stability to DNA and is replaced by uracil in RNA.



(Thymine)

166. Answer (1)

Hint: Heteropolymers constitute 10-15% of the total cellular mass.

Sol.: Proteins constitute 10-15% of the total cellular mass. Proteins are polymers of amino acids joined together by peptide bonds.

Component	% of the total cellular mass
Water	70-90
Proteins	10-15
Carbohydrates	3
Lipids	2
Nucleic acids	5-7
Ions	1

167. Answer (2)

Hint: Glucose is a hexose sugar.

Sol.: GLUT-4 is the insulin-regulated glucose transporter responsible for facilitating the transport of glucose into the cells.

The one feature common to all those compounds found in the acid-soluble pool is that they have molecular weights ranging from 18 to around 800 daltons approximately.

The acid-insoluble fraction has only four types of organic compounds *i.e.*, proteins, nucleic acids, polysaccharides and lipids. These classes of compounds with the exception of lipids, have molecular weights in the range of ten thousand daltons and above. The chemical compounds which have molecular weights less than one thousand daltons are usually referred to as micromolecules or simply biomolecules.

168. Answer (2)

Hint: Oxygen is required for respiration.

Sol.:

Element	% Weight of	
	Earth's crust	Human body
Hydrogen (H)	0.14	9.5
Carbon (C)	0.03	18.5
Oxygen (O)	46.6	65.0
Nitrogen (N)	very little	3.3
Sulphur (S)	0.03	0.3
Sodium (Na)	2.8	0.2
Calcium (Ca)	3.6	1.5
Magnesium (Mg)	2.1	0.1
Silicon (Si)	27.7	negligible

169. Answer (4)

Hint: Co-factors are crucial for catalytic activity of an enzyme.

Sol.: Catalase – Haem is the prosthetic group.

NADP – Co-enzyme containing vitamin niacin.

Peroxidase – Breaks hydrogen peroxide into water.

Carboxypeptidase – Loses its catalytic activity after removal of its co-factor (Zn^{+2}).

170. Answer (1)

Hint: In the structure of DNA, [A] pairs with [T]

Sol.: According to Chargaff's rule:

A = T = 30% $\rightarrow$ A + T = 60%

Remaining = 40% $\rightarrow$ G + C

$\therefore$ G = C = 20%

171. Answer (4)

Hint: Unsaturated fatty acid has comparatively lower melting point.

Sol.: Saturated fatty acid has higher melting point than unsaturated fatty acid.

Palmitic acid is a saturated fatty acid (without double bond) with 16-carbon atoms while, arachidonic acid is a 20-carbon atoms containing unsaturated fatty acid (with double bond).

The absence of double bonds allow their molecular chains to pack closely together in a straight structure.

This tight packing creates stronger intramolecular forces, requiring more heat to melt them.

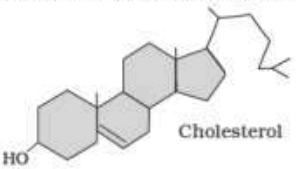
172. Answer (2)

Hint: Found in cell membrane

Sol.: Inorganic catalysts can often withstand extreme conditions.

Physical changes are not catalyzed by enzymes.

Aquaporins are channel-forming membrane proteins that facilitate the rapid transport of water across biological cell membrane.



173. Answer (2)

Hint: Both are equal

Sol.: In the formation of a linear polysaccharide chain with 50 glucose units, 49 glycosidic bonds will be formed.

For a chain containing (n) monosaccharide units:-

Number of water molecules removed will be

$\rightarrow$ n – 1

Number of glycosidic bonds formed will be

$\rightarrow$ n – 1

174. Answer (2)

Hint: Malonate *vs* succinate

Sol.: When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of the enzyme, it is known as competitive inhibitor. Due to its close structural similarity with the substrate, the inhibitor competes with the substrate for the substrate binding site of the enzyme. Consequently, the substrate cannot bind and as a result, the enzyme activity declines. In competitive inhibition, an inhibitor binds to the enzyme's active site, preventing the substrate from binding. Because the substrate and inhibitor directly compete for the same site, the apparent (K_m) increases, meaning a higher substrate concentration is required to reach half the maximum velocity (V_{max}).

175. Answer (4)

Hint: It contains a repeating unit of the same sugar.

Sol.: Chitin is a nitrogen-containing linear polysaccharide and the second most abundant organic substance. It is composed of N-acetylglucosamine.

It forms the exoskeleton of arthropods and cell wall of fungi. Chitin is a homopolymer.

176. Answer (3)

Hint: Haemoglobin has this kind of protein structure

Sol.: Some proteins are an assembly of more than one polypeptide or subunits. The manner in which these individual folded polypeptides or subunits are arranged with respect to each other (*e.g.*, linear string of spheres, spheres arranged one upon each other in the form of a cube or plate, *etc.*) is the architecture of a protein otherwise called the quaternary structure of a protein.

Quaternary protein is stabilized by four primary forces: Non-covalent interactions, hydrogen bonds, ionic bonds, and disulfide bridges.

177. Answer (2)

Hint: Proteins are heteropolymers.

Sol.: Proteins are not homopolymers. Ribulose bisphosphate Carboxylase-Oxygenase (RuBisCO) is the most abundant protein in the whole of the biosphere. Antibodies are made up of polypeptide chains.

Starch and cellulose are homopolymers.

178. Answer (3)

Hint: Cellulose is a homopolymer.

Sol.: Cellulose is a polymeric polysaccharide consisting of only one type of monosaccharide *i.e.*, glucose. Starch forms helical secondary structures. Starch can hold I_2 molecules in its helical portion. Cellulose does not contain complex helices and hence cannot hold I_2.

A nucleotide has three chemically distinct moiety. One is a heterocyclic compound, the second is a monosaccharide and the third is phosphoric acid.

The heterocyclic compounds in nucleic acids are the nitrogenous bases and the sugar molecule. The sugar found in polynucleotides is either ribose (a monosaccharide pentose) or 2′ deoxyribose.

179. Answer (3)

Hint: Inulin contains fruit sugar.

Sol.: Inulin is a natural polymer composed of fructose (fruit sugar), hence called fructans. Its monomeric unit is connected *via* glycosidic bonds. It acts as a reserve carbohydrate in plants, such as Dahlia roots and is commonly classified as a dietary fibre.

180. Answer (4)

Hint: Also called phosphodiester bond

Sol.: Ester bond in DNA, specifically phosphodiester bond, form the covalent backbone connecting adjacent nucleotides. A single phosphate group creates two ester linkages, connecting the 3′ carbon of one deoxyribose sugar to 5′ carbon of the next, providing structural stability to the double helix.

❑ ❑ ❑

All India Aakash Test Series for NEET - 2028

TEST - 1 (Code-D)

Click here for Code-C Sol.

Test Date : 30/08/2026

ANSWERS

1. (3)	37. (3)	73. (2)	109. (2)	145. (4)
2. (1)	38. (3)	74. (1)	110. (3)	146. (1)
3. (1)	39. (4)	75. (1)	111. (2)	147. (4)
4. (3)	40. (2)	76. (1)	112. (2)	148. (2)
5. (4)	41. (3)	77. (4)	113. (3)	149. (2)
6. (4)	42. (1)	78. (2)	114. (4)	150. (1)
7. (1)	43. (4)	79. (1)	115. (4)	151. (3)
8. (3)	44. (2)	80. (1)	116. (4)	152. (3)
9. (2)	45. (3)	81. (2)	117. (2)	153. (2)
10. (1)	46. (2)	82. (3)	118. (3)	154. (3)
11. (4)	47. (1)	83. (3)	119. (2)	155. (2)
12. (4)	48. (1)	84. (4)	120. (1)	156. (2)
13. (1)	49. (2)	85. (3)	121. (2)	157. (1)
14. (4)	50. (4)	86. (2)	122. (2)	158. (1)
15. (3)	51. (1)	87. (2)	123. (3)	159. (3)
16. (3)	52. (4)	88. (1)	124. (3)	160. (2)
17. (1)	53. (3)	89. (4)	125. (3)	161. (3)
18. (4)	54. (1)	90. (2)	126. (3)	162. (1)
19. (1)	55. (3)	91. (3)	127. (3)	163. (3)
20. (3)	56. (4)	92. (2)	128. (4)	164. (3)
21. (4)	57. (2)	93. (4)	129. (3)	165. (1)
22. (1)	58. (3)	94. (4)	130. (1)	166. (4)
23. (4)	59. (2)	95. (4)	131. (2)	167. (2)
24. (1)	60. (4)	96. (3)	132. (3)	168. (2)
25. (2)	61. (4)	97. (2)	133. (2)	169. (3)
26. (2)	62. (2)	98. (1)	134. (3)	170. (2)
27. (2)	63. (1)	99. (3)	135. (4)	171. (2)
28. (4)	64. (1)	100. (4)	136. (4)	172. (2)
29. (4)	65. (2)	101. (1)	137. (3)	173. (3)
30. (2)	66. (1)	102. (2)	138. (3)	174. (3)
31. (3)	67. (1)	103. (1)	139. (2)	175. (4)
32. (3)	68. (1)	104. (4)	140. (3)	176. (1)
33. (3)	69. (2)	105. (4)	141. (4)	177. (3)
34. (3)	70. (1)	106. (3)	142. (2)	178. (4)
35. (1)	71. (4)	107. (3)	143. (2)	179. (1)
36. (3)	72. (1)	108. (4)	144. (2)	180. (2)

HINTS & SOLUTIONS

[PHYSICS]

1. Answer (3)

Hint: Stopping distance $S = \frac{u^2}{2a}$

Sol.: Distance travelled during reaction time:

$S_1 = V(\Delta t) = 30 \times 0.5 = 15$ m

Stopping distance under braking:

$$S_2 = \frac{u^2}{2a} = \frac{(30)^2}{2\times 5} = 90 \text{ m}$$

Total stopping distance

$S = S_1 + S_2 = 15 + 90 = 105$ m

2. Answer (1)

Hint & Sol.: Area under velocity time curve gives displacement. Slope of velocity time curve gives acceleration.

Area under acceleration time curve gives change in velocity.

3. Answer (1)

Hint: There are 7 fundamental physical quantities

Sol.: Electric current $\rightarrow$ Base quantity

Plane angle $\rightarrow$ Supplementary, derived quantity

Force $\rightarrow$ Derived quantity

4. Answer (3)

Hint: Use principle of homogeneity

Sol.: $[At^2] = [v]$

$$[A] = \left[\frac{v}{t^2}\right] = \frac{[LT^{-1}]}{[T^2]} = [LT^{-3}]$$

$$\left[\frac{Bt}{C+t}\right] = [v] = [LT^{-1}]$$

$[B] = [LT^{-1}]$ and $[C] = [T]$

$\therefore [ABC] = [LT^{-3}]\,[LT^{-1}]\,[T]$

$[ABC] = [L^2T^{-3}] = [M^0L^2T^{-3}]$

5. Answer (4)

Hint: Acceleration $a = \frac{dv}{dt}$

Sol.: $a = \frac{dv}{dt}$

$a = 0 - 3t^2$

$a = -3t^2$ is non-uniform

Negative sign indicates that car is undergoing retardation.

6. Answer (4)

Hint & Sol.: Some of the sources of systematic error are as follows:

(1) Instrumental error

(2) Imperfection in experimental technique or procedure

(3) Personal error

7. Answer (1)

Hint: Slope of displacement time curve gives velocity.

Sol.: Ratio of velocity

$$= \frac{\tan 53^\circ}{\tan 37^\circ} = \frac{\frac{4}{3}}{\frac{3}{4}} = \frac{16}{9}$$

8. Answer (3)

Hint: Instantaneous acceleration is the rate of change of velocity

Sol.: For uniform acceleration, velocity is given as $V^2 = u^2 + 2aS$

Graph between $(V - S)$ will be parabola

- Both velocity and acceleration can change within a time interval.

9. Answer (2)

Hint: Use rules of determining number of significant figure

Sol.: According to rules of significant figures, insignificant zeros are three.

10. Answer (1)

Hint: Use principle of homogeneity

Sol.: $\left[\frac{a}{V^2}\right] = [P]$

$[a] = [PV^2]$ ...(i)

$[b] = [V]$ ...(ii)

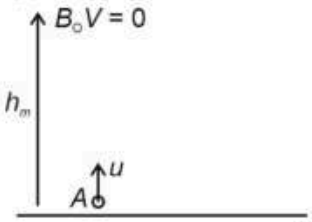
$$\left[\frac{a}{b^2}\right] = \left[\frac{PV^2}{V^2}\right] = [P] = [ML^{-1}T^{-2}]$$

$$[\text{Energy density}] = \left[\frac{U}{V}\right] = \frac{[ML^2T^{-2}]}{[L^3]} = [ML^{-1}T^{-2}]$$

11. Answer (4)

Hint: Average speed $V_{avg} = \frac{\text{Total distance}}{\text{Total time taken}}$

Sol.:



Maximum height $h_m = \frac{u^2}{2g}$

Time of flight $T = \frac{2u}{g}$

For complete journey:

Total distance $s = 2h_m$

Total time taken $T = \frac{2u}{g}$

$$\therefore \quad V_{avg} = \frac{2h_m}{2\,u/g} = \frac{2\left(u^2/2g\right)}{2(u/g)}$$

$$V_{avg} = \frac{u}{2} = \frac{20}{2} = 10 \text{ m/s}$$

Average acceleration $a_{avg} = \frac{V_f - V_i}{T}$

$$= \frac{u-(-u)}{\frac{2u}{g}} = g = 10 \text{ m/s}^2$$

12. Answer (4)

Hint: Use equation of motion $v^2 = u^2 - 2as$ for retardation a

Sol.: For first part:

$$V^2 = u^2 - 2as$$

$$\left(\frac{V_0}{2}\right)^2 = V_0^2 - 2a(36)$$

$$\frac{3V_0^2}{4} = 72a \qquad \ldots(i)$$

For second part:

$$0^2 = \left(\frac{V_0}{2}\right)^2 - 2as'$$

$$\frac{V_0^2}{4} = 2as' \qquad \ldots(ii)$$

From (i) and (ii)

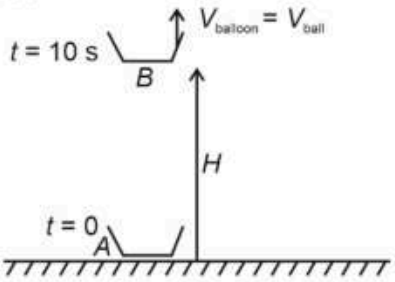
$3(2as') = 72a$

$s' = 12$ cm

13. Answer (1)

Hint: When ball is dropped from balloon, $V_{balloon} = V_{ball}$

Sol.:



For A to B :

$V_{Balloon} = 0 + at$

$= 2 \times 10$

$= 20$ m/s

By relative velocity

$V_{Balloon} = V_{ball} = 20$ m/s

Further height risen by ball,

$$h_{max} = \frac{V_{ball}^2}{2g} = \frac{(20)^2}{2\times 10} = 20 \text{ m}$$

Total height from ground, $H' = h_{max} + H$

$H' = 20 + H \qquad \ldots(i)$

For balloon :

$$H = 0 + \frac{a}{2}t^2$$

$$H = \frac{2}{2}(10)^2 = 100 \text{ m}$$

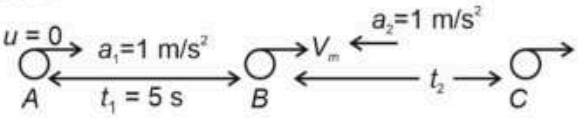
From (i),

$H' = 20 + 100 = 120$ m

14. Answer (4)

Hint: Use equations of motion

Sol.:



For $A \rightarrow B$:

$V_m = u + at_1$

$V_m = 0 + 1 \times 5$

$V_m = 5$ m/s

For $B \rightarrow C$:

$V = u + at$

$0 = 5 - 1 \times t_2$

$t_2 = 5$ s

For $A \rightarrow C$:

$S_{AC} = S_{AB} + S_{BC}$

$$S_{AC} = 0 + \frac{a_1 t_1^2}{2} + V_m t_2 - \frac{a_2 t_2^2}{2}$$

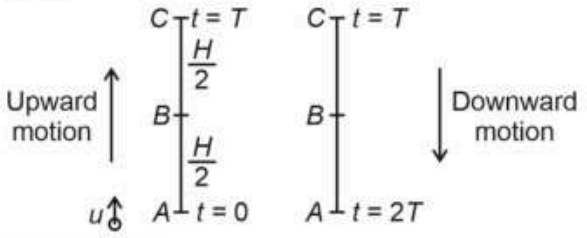
$$S_{AC} = 0 + \frac{1}{2}(5)^2 + 5\times(5) - \frac{1}{2}(5)^2$$

S_{AC} = 25 m

15. Answer (3)

Hint: Use symmetry between upward and downward motion

Sol.:



By symmetry

Time taken in upward motion is same as in downward motion between two given points

Time taken from A to B = Time from B to A = T_1

Time taken from B to C = Time from C to B = T_2

From $C \rightarrow A$: time taken = T

$$H = ut + \frac{a}{2}t^2$$

$$H = 0 + \frac{g}{2}t^2$$

$$T = t = \sqrt{\frac{2H}{g}} = T_2 + T_1$$

From $C \rightarrow B$:

$$t_{C\rightarrow B} = T_2 = \sqrt{\frac{2\left(\frac{H}{2}\right)}{g}} = \frac{T}{\sqrt{2}}$$

From $B \rightarrow A$:

$$t_{B\rightarrow A} = T_1 = T - \frac{T}{\sqrt{2}} = \frac{\left(\sqrt{2}-1\right)T}{\sqrt{2}} = \left(\sqrt{2}-1\right)T_2$$

16. Answer (3)

Hint: Use of equation of motion

Sol.: Acceleration $a = -g$ = constant

- Velocity $v = u - gt$
- Distance $S_t = u - \frac{g}{2}(2t-1)$

It indicates that distance travelled in equal time interval decreases.

17. Answer (1)

Hint: Acceleration $a = v\left(\frac{dv}{ds}\right)$

Sol.: $v^2 = s^2 + 2s + 4$

$$\frac{dv^2}{ds} = \frac{d}{ds}\left[s^2 + 2s + 4\right]$$

$$\frac{2vdv}{ds} = \left[2s + 2\right]$$

$2a = 2s + 2$

$a = s + 1$

$a = 3 + 1$

$a = 4$ m/s^2

18. Answer (4)

Hint: Displacement $\Delta x = \int v dt$

Sol.: Velocity $v = \alpha t - \beta t^2 = 0$

t = 0 or $t = \frac{\alpha}{\beta}$

$$\frac{dx}{dt} = v = \alpha t - \beta t^2$$

$$\int_{x_i}^{x_f} dx = \int_0^{\alpha/\beta} \alpha t dt - \int_0^{\alpha/\beta} \beta t^2 dt$$

$$x_f - x_i = \alpha\left[\frac{t^2}{2}\right]_0^{\alpha/\beta} - \frac{\beta}{3}\left[t^3\right]_0^{\alpha/\beta}$$

(Displacement) $d = \Delta x = \frac{\alpha^3}{6\beta^2}$

19. Answer (1)

Hint: Volume $V = \pi r^2 L$

$$\frac{\Delta V}{V} = \frac{2\Delta r}{r} + \frac{\Delta L}{L}$$

Sol.: $\frac{\Delta V}{V}\times 100 = \frac{2\times 0.01}{0.5}\times 100 + \frac{0.05\times 100}{10}$

$$\frac{\Delta V}{V}\times 100 = 4\% + 0.5\% = 4.5\%$$

Density of wire $d = \frac{m}{V}$

$$\frac{\Delta d}{d} = \frac{\Delta m}{m} + \frac{\Delta V}{V}$$

$$\frac{\Delta d}{d}\times 100 = \frac{0.003}{0.6}\times 100 + 4.5\%$$

$$\therefore \frac{\Delta d}{d}\times 100 = 0.5\% + 4.5\% = 5\%$$

20. Answer (3)

Hint & Sol.: $y = 3t^2 + 2t$

$$\frac{dy}{dt} = 6t + 2 = 8$$

21. Answer (4)

Hint: Average speed = $\frac{\text{Total distance}}{\text{Total time}}$

Sol.: $v_{avg} = \frac{2d}{\frac{d}{u}+\frac{d}{3u}} = \frac{2}{\frac{1}{u}\left[1+\frac{1}{3}\right]} = \frac{3u}{2}$

22. Answer (1)

Hint: Use $\Delta v = \int a dt$ and $\Delta x = \int v dt$

Sol.: Acceleration is given

$$a = \frac{dv}{dt} = 6t + 2$$

$$\int_0^v dv = \int a dt = \int_0^t 6t dt + \int_0^t 2dt$$

$$v = 6\left[\frac{t^2}{2}\right] + 2t$$

$$\frac{dx}{dt} = v = 3t^2 + 2t$$

$$\int_{x_i}^{x_f} dx = \int v dt = \int_0^2 3t^2 dt + \int_0^2 2t dt$$

$$x_f - x_i = 3\left[\frac{t^3}{3}\right]_0^2 + 2\left[\frac{t^2}{2}\right]_0^2$$

$\Delta x = 8 + 4$

$\Delta x = 12$ m

23. Answer (4)

Hint: Least count of screw gauge,

$$LC = \frac{\text{Pitch}}{\text{No. of circular scale division}}$$

Sol.: Initial least count

$$LC_1 = \frac{\text{Pitch } (P)}{\text{No. of circular scale division } (N)}$$

$$LC_1 = \frac{P}{N}$$

Final least count $LC_2 = \frac{P + \frac{75}{100}P}{N - \frac{50}{100}N}$

$$LC_2 = \frac{\frac{7}{4}P}{\frac{N}{2}} = \frac{7}{2}\left[\frac{P}{N}\right]$$

$$LC_2 = \frac{7}{2}LC_1$$

$$LC_2 = \frac{7}{2} \times 0.01 \text{ mm}$$

$$LC_2 = 35 \times 10^{-3} \text{ mm}$$

24. Answer (1)

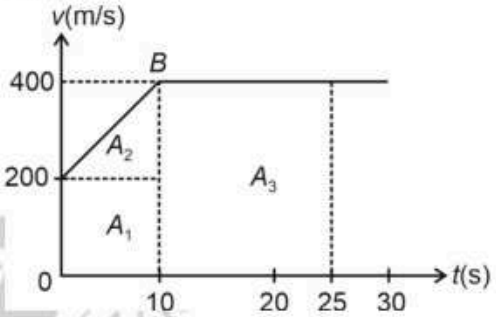
Hint & Sol.: Dimensions of change in energy will be equal to that of energy.

[Energy] = $[ML^2T^{-2}]$

25. Answer (2)

Hint: Displacement = Area under ($v - t$) graph

Sol.:



Distance (S) = $A_1 + A_2 + A_3$

$$S = 200 \times 10 + \frac{1}{2} \times 10 \times 200 + (25 - 10) \times 400$$

$S = 2000 + 1000 + 6000 = 9000$ m

$\therefore S = 9$ km

26. Answer (2)

Hint: Acceleration $a = \frac{dv}{dt}$

Sol.: $a = -K\sqrt{v}$

$$\frac{dv}{dt} = -K\sqrt{v}$$

$$\int_u^0 \frac{dv}{\sqrt{v}} = -\int_0^T K dt$$

$$\left[2\sqrt{v}\right]_u^0 = -KT$$

$$-2\sqrt{u} = -KT$$

$$T = \frac{2\sqrt{u}}{K}$$

27. Answer (2)

Hint: Fractional error $\frac{\Delta g}{g}=\frac{\Delta l}{l}+\frac{2\Delta T}{T}$

Sol.: Time period of a simple pendulum

$T=2\pi\sqrt{\frac{l}{g}}$

$T^2=4\pi^2\frac{l}{g}$

$g=4\pi^2\frac{l}{T^2}$

$\frac{\Delta g}{g}=\frac{\Delta l}{l}+\frac{2\Delta T}{T}$

$\frac{\Delta g}{g}=\frac{0.1}{40}+2\times\frac{\Delta t}{t}$

$\left[\because\ t=nT,\frac{\Delta t}{t}=\frac{n\Delta T}{nT}=\frac{\Delta T}{T}\right]$

$\frac{\Delta g}{g}\times100=\left[\frac{0.1}{40}+2\times\frac{0.5}{80}\right]\times100=1.5\%$

28. Answer (4)

Hint: Observed reading (R) = $MSR + n \times VC$

Sol.: $R = 2.4 + 6 \times 0.01 = 2.46$ cm

Zero correction

= –(Zero error) = –(–0.04 cm) = 0.04 cm

Correct reading (R')

R' = Observed Reading + Zero correction

$R' = 2.46 + 0.04 = 2.50$ cm

29. Answer (4)

Hint: When physical quantities are added, there should be as many decimal places retained in the final result as there are in the number with the least decimal places.

Sol.: 0.1 + 0.01 + 0.001 = 0.111 cm $\approx$ 0.1 cm

30. Answer (2)

Hint: Use $n_1u_1 = n_2u_2$

Sol.: In one system

Force has dimensions $\rightarrow \left[M_1L_1T_1^{-2}\right]$

Dimension of force in new system $\rightarrow \left[M_2L_2T_2^{-2}\right]$

$\because\quad n_2u_2=n_1u_1$

$n_2=\frac{n_1u_1}{u_2}=\frac{10\left[M_1L_1T_1^{-2}\right]}{\left[M_2L_2T_2^{-2}\right]}$

$n_2=10\left[\frac{M_1}{M_2}\right]\left[\frac{L_1}{L_2}\right]\left[\frac{T_1}{T_2}\right]^{-2}$

$n_2=10\left[\frac{1}{\alpha}\right]\left[\frac{1}{\beta}\right]\left[\frac{1}{\gamma}\right]^{-2}$

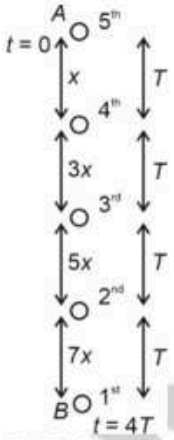
$n_2=10\alpha^{-1}\beta^{-1}\gamma^2$

31. Answer (3)

Hint: By Galileo's ratio

$x_1 : x_2 : x_3 = 1 : 3 : 5$

Sol.:



By Galileo's ratio of odd numbers

$S_{2nd} - S_{3rd} = 5x$

$35 = 5x$

$x = 7$ m ...(i)

From A to B

$S=ut+\frac{a}{2}t^2$

$16x=0+\frac{g}{2}(4T)^2$

$x=\frac{g}{2}T^2$ ...(ii)

From (i) and (ii)

$7=\frac{g}{2}T^2$

$T=\sqrt{\frac{7}{5}}$ s

32. Answer (3)

Hint: Relative error is given as

$\frac{\Delta Z}{Z}=\frac{\Delta p}{p}+\frac{2\Delta q}{q}+\frac{3\Delta r}{r}+\frac{1}{2}\frac{\Delta s}{s}$

Sol.: $\frac{\Delta Z}{Z}\times100$

$= 1\% + 2 \times 2\% + 3 \times 3\% + \frac{1}{2} \times 2\%$

$\frac{\Delta Z}{Z}\times100 = 1 + 4 + 9 + 1 = 15\%$

33. Answer (3)

Hint: Use principle of homogeneity

Sol.: $[A\sin t] = [x(t)]$

$[A][M^0L^0T^0] = [L]$

$[A] = [L]$

Similarly

$[B] = [L]$ and $[D] = [L]$

$[Ct^2] = [x]$

$[Ct^2] = [L]$

$[C] = [LT^{-2}]$

$\therefore \quad \left[\frac{ABC}{D}\right] = \frac{[L][L][LT^{-2}]}{[L]}$

$\left[\frac{ABC}{D}\right] = [L^2T^{-2}] = [M^0L^2T^{-2}]$

34. Answer (3)

Hint: Acceleration $\vec{a} = \frac{d\vec{v}}{dt}$

Sol.: Acceleration is due to change in velocity

- A particle can be momentarily at rest but still experience an ongoing change in velocity or acceleration.
- An object can also accelerate while moving at a constant speed due to change in direction of motion.

35. Answer (1)

Hint: Average acceleration

$a_{avg} = \frac{V_f - V_i}{\Delta t}$

Sol.: $a_{avg} = \frac{\left(90 \times \frac{5}{18}\right) - 0}{10}$

$a_{avg} = \frac{25}{10} = 2.5\ m/s^2$

36. Answer (3)

Hint: Surface Tension $T = \frac{F}{L}$

and Impulse $I = \Delta p$

Sol.: $[T] = \left[\frac{F}{L}\right] = \frac{[MLT^{-2}]}{[L]} = [MT^{-2}]$

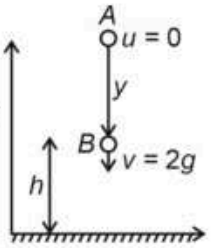
$[I] = [\Delta p] = [MLT^{-1}]$

Dimensions of $T \neq$ dimensions of I.

37. Answer (3)

Hint: Use $v^2 = u^2 + 2as$

Sol.:



From A to B

$v^2 = u^2 + 2as$

$(2g)^2 = 0 + 2 \times 10 \times y$

$y = 20$ m

Height from ground $h = H - y = 60 - 20 = 40$ m

38. Answer (3)

Hint: Average velocity $V_m = \frac{\text{Total displacement}}{\text{Total time}}$

Sol.: $V_m = \frac{d + 1.5d}{t_1 + t_2}$

$\frac{50}{7} = \frac{2.5d}{\left(\frac{d}{5} + \frac{1.5d}{V_0}\right)}$

$V_0 = 10$ m/s

39. Answer (4)

Hint: Acceleration $A = \frac{dv}{dt} = \frac{d^2x}{dt^2}$

Sol.: $x = a\left(t^2 - 2\right) + b(t-2)^2$

Velocity $v = \frac{dx}{dt} = a(2t - 0) + 2b(t-2)$

$v = 2at + 2bt - 4b$

Acceleration $A = \frac{dv}{dt} = 2a + 2b - 0$

$A = 2(a + b)$

40. Answer (2)

Hint: Relative velocity

$V_{A/B} = V_A + V_B$ [Due to opposite direction]

Sol.: $V_{A/B} = 30 + 20$

$V_{A/B} = 50$ m/s

Time of passing $T = \frac{s_{A/B}}{V_{A/B}} = \frac{100 + 150}{50}$

$T = 5$ s

41. Answer (3)

Hint: Velocity of ball w.r.t. elevator

$V_{b/e} = V_b - V_e$

Sol.: $V_b = V_{b/e} + V_e$

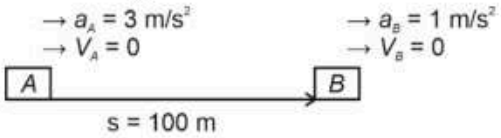
$V_b = 10 + 5$

$V_b = 15$ m/s

42. Answer (1)

Hint: Relative displacement

$$s_{A/B} = u_{A/B}t + \frac{a_{A/B}}{2}t^2$$



Sol.:

Relative initial velocity:

$u_{A/B} = u_A - u_B = 0$

Relative acceleration:

$a_{A/B} = a_A - a_B = 3 - 1 = 2$ m/s²

By equation of motion,

$$s_{A/B} = u_{A/B}t + \frac{a_{A/B}}{2}t^2$$

$$100 = 0 + \frac{2}{2}t^2$$

t = 10 s

43. Answer (4)

Hint: Displacement $S = ut - \frac{g}{2}t^2$

Sol.: If the balls meet at time t

Then first ball will take t seconds while second ball will take $(t-2)$ seconds and they will meet at same height $\rightarrow h_1 = h_2$

$$40t - \frac{g}{2}t^2 = 40(t-2) - \frac{g}{2}(t-2)^2$$

$$40t - \frac{g}{2}t^2 = 40t - 80 - \frac{g}{2}\left(t^2 + 4 - 4t\right)$$

$$40t - \frac{g}{2}t^2 = 40t - 80 - \frac{g}{2}t^2 - 2g + 2gt$$

$0 = -80 - 2 \times 10 + 2 \times 10 \times t$

$$\frac{100}{20} = t$$

t = 5 s

44. Answer (2)

Hint: Least count of screw gauge

$$= \frac{\text{Pitch}}{\text{Number of circular scale division}}$$

Sol.: Least count $= \frac{0.75}{15} = 0.05$ mm

Area of sheet $A = Lb$

$$\frac{\Delta A}{A} = \frac{\Delta L}{L} + \frac{\Delta b}{b}$$

$$\frac{\Delta A}{A} = \frac{0.05}{5} + \frac{0.05}{2.5}$$

$$\frac{\Delta A}{A} = 0.01 + 0.02$$

$$\frac{\Delta A}{A} = 0.03$$

45. Answer (3)

Hint: If a body is dropped from a certain height then, time taken to reach ground, $t = \sqrt{\frac{2h}{g}}$.

Sol.: $\frac{t_1}{t_2} = \sqrt{\frac{h_1}{h_2}} = \sqrt{\frac{16}{25}} = \frac{4}{5}$

[CHEMISTRY]

46. Answer (2)

Hint: Cathode rays consists of negatively charged particle called electrons.

Sol.: The characteristics of cathode rays do not depend upon the material of electrodes and the nature of the gas present in the cathode ray tube.

47. Answer (1)

Hint: Mass number = Total number of neutrons + Total number of protons

Sol.: Number of electrons = X

Number of neutrons = X + 0.304X

= 1.304X

Number of protons = X + 3

56 = X + 3 + 1.304X

Number of protons = 26

So correct symbol is $^{56}_{26}X^{3+}$

48. Answer (1)

Hint: For Lyman series $n_i = 1$

For Balmer series $n_i = 2$

Sol.: $\left(\frac{1}{\lambda_1}\right)_{\text{Lyman}} = R_H \times (1)^2\left[\frac{1}{1} - \frac{1}{4}\right]$

$$\left(\frac{1}{\lambda_1}\right)_{\text{Lyman}} = \frac{3R_H}{4} \quad \ldots(1)$$

$$\left(\frac{1}{\lambda_2}\right)_{Balmer} = R_H \times (1)^2\left[\frac{1}{4}-\frac{1}{16}\right]$$

$$\left(\frac{1}{\lambda_2}\right)_{Balmer} = \frac{12R_H}{64} \quad ...(2)$$

Equation (1) is divided by equation (2)

$$\frac{\frac{1}{\lambda_1}}{\frac{1}{\lambda_2}} = \frac{\frac{3R_H}{4}}{\frac{12R_H}{64}}$$

$$\frac{\lambda_2}{\lambda_1} = \frac{64\times 3}{12\times 4}$$

$$\frac{\lambda_2}{x} = 4$$

49. Answer (2)

Hint: Total number of spectral lines $= \frac{\Delta n(\Delta n+1)}{2}$

Sol.: 5th excited state, n = 6
1st excited state, n = 2

Total spectral lines = $\frac{4\times 5}{2} = 10$

50. Answer (4)

Hint: According to Heisenberg's uncertainty principle, $\Delta x \times \Delta p \geq \frac{h}{4\pi}$

Sol.: $\Delta x = 2\Delta p$

$$\Delta p \times 2\Delta p = \frac{h}{4\pi}$$

$$2m^2\Delta v^2 = \frac{h}{4\pi}$$

$$\Delta v = \frac{1}{2m}\sqrt{\frac{h}{2\pi}}$$

$$\frac{r_3}{r_4} = \frac{a_0 \times \frac{9}{3}}{a_0 \times \frac{16}{4}}$$

$$\frac{x}{r_4} = \frac{3}{4}$$

$$r_4 = \frac{4x}{3}$$

51. Answer (1)

Hint: $\Delta E = R_H Z^2\left[\frac{1}{n_i^2}-\frac{1}{n_f^2}\right]$

Sol.: For n = 5 → 4

$$\Delta E = R_H\left[\frac{1}{16}-\frac{1}{25}\right]$$

$$\Delta E = \frac{R_H \times 9}{16\times 25}$$

For n = 6 → 4

$$\Delta E = R_H\left[\frac{1}{16}-\frac{1}{36}\right]$$

$$\Delta E = \frac{20R_H}{16\times 36}$$

For n = 5 → 3

$$\Delta E = R_H\left[\frac{1}{9}-\frac{1}{25}\right] = \frac{16R_H}{9\times 25}$$

For n = 6 → 1

$$\Delta E = R_H\left[\frac{1}{1}-\frac{1}{36}\right] = \frac{35R_H}{36}$$

52. Answer (4)

Hint: Isoelectronic species have same number of electrons.

Sol.: NO^+, CO and CN^- have 14 electrons

53. Answer (3)

Hint: The general electronic configuration of Cu atom is $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^6$, $4s^1$, $3d^{10}$

Sol.: 4s and 3p orbitals have (n + l) value four. So total 7 electrons are present.

54. Answer (1)

Hint: Hund's rule states that "pairing of electrons in the orbitals belonging to the same subshell does not take place until each orbital belonging to that subshell has got one electron each."

Sol.: ⇅ ⇅ | 1 |

It does not follow Hund's rule of maximum multiplicity.

55. Answer (3)

Hint: $E = h\nu_0 + KE$

Sol.: $\frac{hc}{\lambda} = w_0 + 1.74\times 10^5$

$$w_0 = \frac{6.6\times 10^{-34}\times 3\times 10^8\times 6\times 10^{23}}{300\times 10^{-9}} - 1.74\times 10^5$$

$$w_0 = 2.2\times 10^5 \text{ J}$$

56. Answer (4)

Hint: When electromagnetic radiations are arranged in their increasing order of wavelength, then the spectrum which is obtained is called the electromagnetic spectrum.

Sol.: Correct order of wavelength of electromagnetic radiations

= IR > UV > X rays > γ rays

57. Answer (2)

Hint: The lower the value of (n + l) for an orbital, the lower is its energy. If two orbitals have the same value of (n + l), then the orbital with lower value of n will have the lower energy.

Sol.:

Orbital	(n + l) value
5*d*	7
5*f*	8
7*s*	7
6*p*	7

58. Answer (3)

Hint: de Broglie wavelength $(\lambda) = \frac{h}{mv}$

Sol.: $\lambda = \frac{6.626 \times 10^{-34}\ \text{J s}}{10 \times 10^{-3} \times 10^{-3} \times 100}$

$\lambda = 6.626 \times 10^{-31}\ \text{m}$

59. Answer (2)

Hint: The energy of an electron in a hydrogen atom is determined solely by the principal quantum number.

Sol.: The correct order of energies of atomic orbitals for hydrogen atom is

$1s < 2s = 2p < 3s = 3p = 3d < 4s$

60. Answer (4)

Hint: d_{z^2} and $d_{x^2-y^2}$ orbitals have electron density along the axes.

Sol.: Two electrons occupying the same orbital are distinguished by spin quantum number.

61. Answer (4)

Hint: The maximum number of electrons in the shell with principal quantum number n is equal to $2n^2$

Sol.:

Work function (w_0)	Metal
2.42	Li
3.7	Mg

Balmer spectral line lies in visible region while Paschen spectral line lies in infrared region.

62. Answer (2)

Hint: The value of m_l = –l to +l including zero

Sol.: For n = 4

l = 0, 1, 2, 3

63. Answer (1)

Hint: de Broglie wavelength $(\lambda) = \frac{h}{mv}$

Sol.: $\lambda = \frac{2\pi r}{n}$ ...(1)

$r_4 = a_0 \times \frac{16}{1}$ ...(2)

Putting the value of r_4 in equation (1)

$\lambda = \frac{2\pi \times 16a_0}{4}$

$\lambda = 8\pi a_0$

64. Answer (1)

Hint: Radial nodes = (n – l – 1)

Sol.:

Orbital	Radial node(s)
4*p*	2
4*d*	1
5*s*	4
4*f*	0

65. Answer (2)

Hint: Orbital angular momentum $= \hbar\sqrt{l(l+1)}$

Sol.: For p orbital, l = 1

Orbital angular momentum $= \hbar\sqrt{1(1+1)} = \hbar\sqrt{2}$

66. Answer (1)

Hint: $\frac{1}{\lambda} = R_H Z^2 \left[\frac{1}{n_i^2} - \frac{1}{n_f^2}\right]$

Sol.: $\frac{1}{\lambda_1} = R_H \left[\frac{1}{9} - \frac{1}{16}\right]$

$\frac{1}{\lambda_1} = R_H \times \frac{7}{16 \times 9}$ ...(1)

$\frac{1}{\lambda_2} = R_H \left[\frac{1}{16} - \frac{1}{25}\right]$

$\frac{1}{\lambda_2} = R_H \times \frac{9}{16 \times 25}$ ...(2)

Dividing equation (1) by equation (2)

$\frac{\lambda_2}{\lambda_1} = \frac{\frac{7R_H}{16 \times 9}}{\frac{9R_H}{16 \times 25}}$

$$\frac{\lambda_2}{\lambda_1}=\frac{7\times16\times25}{16\times9\times9}$$

$$\frac{\lambda_2}{\lambda_1}=\frac{175}{81}$$

67. Answer (1)

Hint: In photoelectric effect, electrons were ejected when certain metals like potassium, caesium were exposed to a beam of light.

Sol.: The kinetic energy of the ejected electrons increases with increase of frequency of the light used.

68. Answer (1)

Hint: The total number of orbitals associated with the principal quantum number (n) is n^2

Sol.: The degeneracy of the third excited state of H atom will be $(4)^2 \Rightarrow 16$

69. Answer (2)

Hint: Energy in n^{th} orbit for one electron system $= -13.6\times\frac{Z^2}{n^2}$ eV/atom

Sol.: Energy of 3rd excited state of He^+ ion

$(E_4) = -13.6\times\frac{4}{16} = -3.4$ eV/atom

Energy in 2nd excited state of H atom

$(E_3) = -13.6\times\frac{1}{9} = -1.51$ eV/atom

Energy in 4th Bohr orbit of Be^{3+}

$(E_4) = -13.6\times\frac{16}{16} = -13.6$ eV/atom

70. Answer (1)

Hint: Energies of the orbitals in the same subshell decrease with increase in the atomic number (Z_{eff})

Sol.: $E_{2s}(H) > E_{2s}(Li) > E_{2s}(Na) > E_{2s}(K)$

71. Answer (4)

Hint: Weight of hydrated $BaCl_2$ sample - weight of dried sample = weight of H_2O

Sol.: Weight of hydrated $BaCl_2$ = 122 g

Loss in mass = 18 g

Mass of H_2O = 18 g

Moles of $H_2O = \frac{18}{18}$ = 1 mol

Gram molecular mass of $BaCl_2$ = 208 g/mol

Moles of anhydrous $BaCl_2 = \frac{104}{208} = \frac{1}{2} = 0.5$ mol

0.5 moles of $BaCl_2$ combines with = 1 mol of H_2O

1 mole of $BaCl_2$ combines with $= \frac{10}{0.5}$ = 2 mol of H_2O

So correct formula = $BaCl_2\cdot 2H_2O$

72. Answer (1)

Hint: g eq. of metal = g eq. of oxygen

$\frac{68}{E}=\frac{32}{8}$

E = 17 g

Sol.:

	C_3H_8	+ $5O_2$	→ $3CO_2$	+ $4H_2O$
Mole ⇒	0.5	5		
Left ⇒	0	2.5	1.5	2

Mole fraction of $CO_2 \Rightarrow \frac{1.5}{6} \Rightarrow 0.25$

73. Answer (2)

Hint: $Ba(OH)_2 + 2HCl \rightarrow BaCl_2 + 2H_2O$

Sol.: Millimoles of H^+ ion = millimoles of HCl = 2.5 mmol

Millimoles of OH^- ion = 2 × millimoles of $Ba(OH)_2$ = 1.6 mmol

1.6 mmol of OH^- ion react with 1.6 mmol of H^+ ion.

So remaining mmol of H^+ ion = 2.5 – 1.6 = 0.9

$[H^+] = \frac{0.9}{90} = 0.01$ M

74. Answer (1)

Hint: Empirical formula represents the simplest whole number ratio of various atoms present in a compound.

Sol.:

Element	Percentage	Mole	Simplest ratio
C	60%	$\frac{60}{12}=5$	$\frac{5}{2}=2.5\times2=5$
H	12%	$\frac{12}{1}=12$	$\frac{12}{2}=6\times2=12$
N	28%	$\frac{28}{14}=2$	$\frac{2}{2}=1\times2=2$

Empirical formula = $C_5H_{12}N_2$

75. Answer (1)

Hint: $\frac{\text{Mole of } O_2}{\text{Mole of } N_2} = \frac{\frac{2}{32}}{\frac{3}{28}} = \frac{2\times28}{3\times32} = 7:12$

Sol.: In 1 g of O_2 gas, number of atoms

$= \frac{1}{32} \times N_A \times 2 = \frac{N_A}{16}$ atoms

In 1 g of O_3 gas, number of atoms

$= \frac{1}{48} \times N_A \times 3 = \frac{N_A}{16}$ atoms

76. Answer (1)

Hint: Molarity (M) = $\frac{\text{Number of moles of solute}}{\text{Volume of solution (in L)}}$

Sol.: Mole of HNO_3 = 0.5

Mass of HNO_3 = 0.5 × 63 g

70 g HNO_3 is present in 100 g of solution

0.5 × 63 g HNO_3 is present in $\frac{100}{70} \times 0.5 \times 63$

$\Rightarrow$ 45.0 g conc. HNO_3

77. Answer (4)

Hint: Molarity (M) = $\frac{\text{Number of moles of solute}}{\text{Volume of solution (in L)}}$

Sol.: $M = \frac{3.01 \times 10^{22} \times 1000}{6.02 \times 10^{23} \times 200} = 0.25$ M

Molality (m) = $\frac{6.02 \times 10^{23} \times 1000}{6.02 \times 10^{23} \times 500} = 2$ m

78. Answer (2)

Hint & Sol.:

$$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O$$

Moles = 1 mol 2 mol

Here NH_3 is limiting reagent

4 mol NH_3 produces 4 mol NO

1 mol NH_3 produces 1 mol NO = 30 g NO

4 mol NH_3 produces 6 mol H_2O

1 mole NH_3 produces $\frac{6}{4}$ mol H_2O

$\Rightarrow \frac{6}{4} \times 18 \Rightarrow 27$ g H_2O

79. Answer (1)

Hint: Mole fraction of solute

= $\frac{\text{Number of moles of solute}}{\text{Total number of moles}}$

Molality = $\frac{\text{Number of moles of solute}}{\text{Weight of solvent (in kg)}}$

Sol.: 2 m means 2 mole of solute dissolve in 1000 gm of solvent.

Moles of solvent = $\frac{1000}{18} = 55.55$

Mole fraction of solute = $\frac{2}{2 + 55.55} \Rightarrow 0.035$

80. Answer (1)

Hint: $3BaCl_2 + 2Na_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6NaCl$

Sol.: Here $BaCl_2$ is limiting reagent.

3 moles of $BaCl_2$ gives 1 mole of $Ba_3(PO_4)_2$

0.1 mole of $BaCl_2$ gives $\frac{1}{3} \times 0.1 \Rightarrow 0.03$ mol of $Ba_3(PO_4)_2$

81. Answer (2)

Hint: Mole = $\frac{\text{Weight (in g)}}{\text{Atomic mass}}$

Sol.: Mole of P = $\frac{4}{40}$ = 0.1 mol

Mole of Q = $\frac{6.0 \times 10^{23}}{6.0 \times 10^{23}}$ = 1 mol

Here R is limiting reagent

3 moles of R produce 1 mole of PQR_3

0.06 moles of R produce $\frac{1}{3} \times 0.06 \Rightarrow 0.02$ mol

Atomic weight of $PQR_3 = \frac{9.4}{0.02} = 470$

$40 + x + 80 \times 3 = 470$

x = 190 u

82. Answer (3)

Hint: % purity = $\frac{\text{Weight of pure substance}}{\text{Weight of impure sample}} \times 100$

Sol.: $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

Moles of $CO_2(g) = \frac{4.4}{44} = 0.1$ mol

1 mol CO_2 is liberated from 1 mol $CaCO_3$

0.1 mol CO_2 is liberated from 0.1 mol $CaCO_3$

Weight of $CaCO_3$ = 0.1 × 100 = 10 g

% purity = $\frac{10}{15} \times 100 = 66.67\%$

83. Answer (3)

Hint and Sol.: According to Dalton's atomic theory, atoms are neither created nor destroyed in a chemical reaction.

84. Answer (4)

Hint: The prefix used for 10^{-18} multiple is atto

Sol.:

Multiple	Prefix
10^{-12}	pico
10^{12}	tera
10^{18}	exa

85. Answer (3)

Hint and Sol.: Abundance of B-10 $\Rightarrow$ x%

Abundance of B-11 $\Rightarrow$ (100 – x)%

$$10.81 = \frac{10x + 11(100 - x)}{100}$$

$$\frac{10x + 1100 - 11x}{100} = 10.81$$

x = 19%

Abundance of heavier isotope = 100 – 19 = 81%

86. Answer (2)

Hint: For minimum molar mass of enzyme, one zinc atom should be present

Sol.: $0.5 = \frac{1 \times 65}{\text{Minimum molar mass of enzyme}} \times 100$

Minimum molar mass of enzyme = 13000 g mol^{-1}

87. Answer (2)

Hint: According to law of multiple proportions, two elements can combine to form more than one compound, the mass of one element that combines with a fixed mass of the other element, is in the ratio of small whole number.

Sol.:

$$\underset{\frac{9}{90} \Rightarrow 0.1 \text{ mol}}{H_2C_2O_4} + H_2SO_4 \rightarrow H_2O + \underset{0.1 \text{ mol}}{CO_2} + \underset{0.1 \text{ mol}}{CO}$$

$$\underset{\frac{4.6}{46} \Rightarrow 0.1 \text{ mol}}{HCOOH} + H_2SO_4 \rightarrow H_2O + \underset{0.1 \text{ mol}}{CO}$$

Total mass of CO gas = 0.2 × 28 $\Rightarrow$ 5.6 g

Total mass of CO_2 gas = 0.1 × 44 = 4.4 g

Total weight of evolved gases = 5.6+4.4 = 10.0 g

- Total number of nucleons = $\frac{1.6}{16} \times N_A \times 16$

$\Rightarrow 1.6\ N_A$

- Significant figure in 200 eggs is ∞

88. Answer (1)

Hint: Number of mmoles of HCl = $\frac{1}{2}$ × Number of mmoles of $CaCO_3$

Sol.: $CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$

Millimoles of HCl = 0.5 × 50 = 25 mmol

2 mmol of HCl reacts with 1 mmol of $CaCO_3$

25 mmol of HCl reacts with $\frac{1}{2}$ × 25 $\Rightarrow$ 12.5 mmol of $CaCO_3$

Mass of $CaCO_3 = \frac{12.5}{1000} \times 100 = 1.25$ g

89. Answer (4)

Hint: $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$

Sol.:

$$\begin{array}{lccc} & N_2(g) & + 3H_2(g) & \rightarrow 2NH_3(g) \\ \text{Mole} \Rightarrow & 1 & 4 & \end{array}$$

Here $N_2(g)$ is limiting reagent.

1 mole $N_2(g)$ produces 2 moles of $NH_3(g)$

Mass of NH_3 = 2 × 17 $\Rightarrow$ 34 g of NH_3

90. Answer (2)

Hint: Number of atoms = Number of moles × atomicity × N_A

Sol.: Number of atoms in 32 g of

$CH_4 = \frac{32}{16} \times 5 \times N_A = 10N_A$ atoms

Number of atoms in 3.011×10^{23} molecules of

$SO_2 = \frac{3.011 \times 10^{23}}{6.022 \times 10^{23}} \times N_A \times 3 = 1.5N_A$ atoms

Number of atoms in 44.8 L of $CO_2(g)$ at STP

$= \frac{44.8}{22.4} \times N_A \times 3 = 6N_A$ atoms

Number of atoms in 17 g of

$NH_3(g) = \frac{17}{17} \times N_A \times 4 = 4N_A$ atoms

[BOTANY]

91. Answer (3)

Hint: Human RBCs contain approximately 52% proteins in the membrane.

Sol.:

Mitochondria	–	Double membrane bound structure that divide by fission
Mycoplasma	–	0.3 μm in length and lacks cell wall
Viruses	–	Non-cellular, exception to cell theory

92. Answer (2)

Hint: Robert Brown described the nucleus as early as in 1831.

Sol.:

Golgi apparatus	–	First observed in 1898 near the nucleus by Camillo Golgi.
Plasma membrane	–	Fluid mosaic model was proposed in 1972
Ribosomes	–	First observed under electron microscope as dense particles in 1953

93. Answer (4)

Hint: A special membranous structure is the mesosome which is formed by the extensions of plasma membrane into the cell and help in cell wall formation.

Sol.:

Fimbriae	–	Small bristle like fibres, do not play role in motility.
Glycocalyx	–	Protects the cells and also helps in adhesion.
Contractile vacuole	–	Found in *Amoeba*

94. Answer (4)

Hint: Chromatin is found in non-dividing cells nucleus.

Sol.: During different stages of cell division, cells show chromosomes in place of nucleus.

95. Answer (4)

Hint: Plastids are found in almost all plant cells and in euglenoids.

Sol.: Nucleus is present in WBCs. Cilia is absent in prokaryotes. Polysome is found in prokaryotes and synthesises proteins.

96. Answer (3)

Hint: Anthocyanin is a water soluble pigment.

Sol.: Microfilaments help in pseudopodia formation. Cisternae of Golgi apparatus resemble with SER.

97. Answer (2)

Hint: Antonie Von Leeuwenhoek first saw and described a live cell.

Sol.: Rudolf Virchow gave *Omnis cellula-e cellula i.e.* cells arise from pre-existing cells.

98. Answer (1)

Hint: Plastids are characteristics of plant cells.

Sol.: Animal cells have centrioles which are absent in almost all plant cells. Large sap vacuoles are found in plant cells.

99. Answer (3)

Hint: Acrocentric chromosome have sub-terminal centromere.

Sol.: Acrocentric chromosomes appear J-shaped during anaphase.

100. Answer (4)

Hint: Cell wall is the outermost covering of plant cell.

Sol.: Primary cell wall is formed towards outer side of plasma membrane.

101. Answer (1)

Hint: Free ribosomes synthesize non-secretory proteins.

Sol.: Plasma membrane have functions like cell growth, formation of intercellular junctions, secretion, endocytosis and cell division.

102. Answer (2)

Hint: In prokaryotes, chemically complex envelope system is found.

Sol.: Primary lysosome contain inactive hydrolytic enzymes.

In polysome, several ribosomes may attach to a single mRNA.

103. Answer (1)

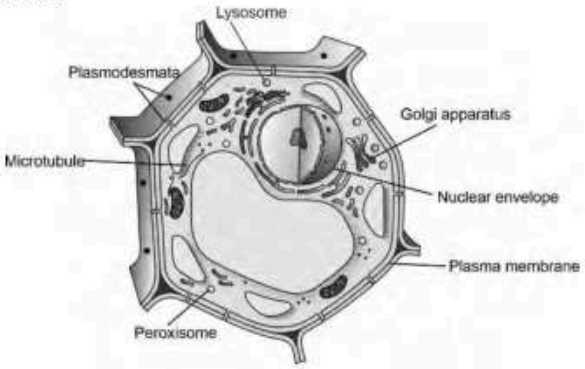
Hint: Endomembrane system include single membrane bound organelles.

Sol.: Carbohydrate synthesis is the function of chloroplast.

104. Answer (4)

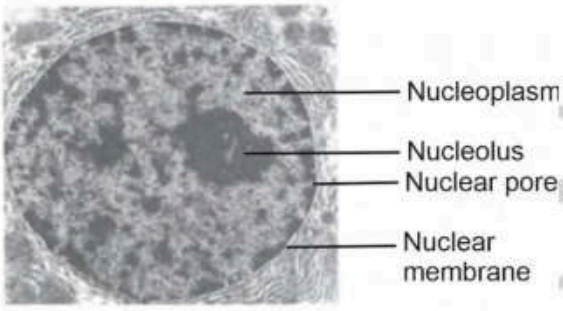
Hint: The given diagram is of plant cell because there is presence of large central vacuole.

Sol.:



105. Answer (4)

Hint:



Structure of nucleus

Sol.: Nucleolus is the site for active synthesis of rRNA. Nuclear pores allow the movement of RNA and proteins in both directions.

106. Answer (3)

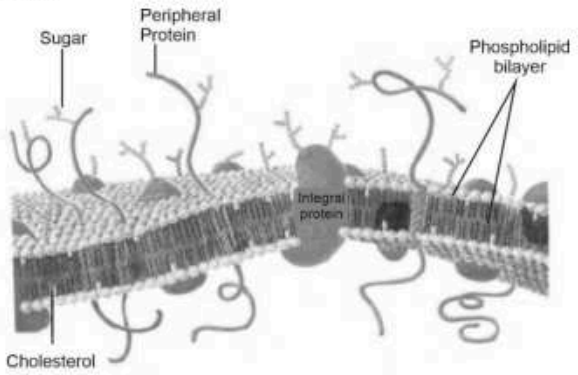
Hint: In a eukaryotic cell, ribosomes are not found in nucleus.

Sol.: Mitochondria does not possess sugar synthesising machinery.

107. Answer (3)

Hint: The given structure is of fluid mosaic model of plasma membrane, proposed by Singer and Nicolson (1972).

Sol.:



Integral protein which run throughout lipid bilayer cannot be removed easily but peripheral proteins can be removed. Lipids perform flip-flop movement.

108. Answer (4)

Sol.: The two ribosomal subunits remain united due to specific concentration of Mg^{++}.

109. Answer (2)

Hint: In centrosome, both the centrioles lie perpendicular to each other in which each has an organisation like cartwheel.

Sol.: Centrosome is non-membrane bound organelle.

110. Answer (3)

Hint: Chloroplast perform various functions like photosynthesis and storage of starch.

Sol.: Mitochondria performs oxidative phosphorylation. RER gives rise to SER.

111. Answer (2)

Hint: Svedberg's unit is indirectly a measure of density and size of ribosomal subunits.

Sol.: Onion peel cell has cell wall as the outer boundary. Prokaryotic ribosomes can be found in cytoplasm.

112. Answer (2)

Hint: Golgi apparatus is responsible for the synthesis of glycoproteins and glycolipids.

Sol.: In fungi, cell wall is made up of chitin. In algae, it is made up of cellulose, galactans mannans and calcium carbonate. In plant, it is made up of cellulose, hemicellulose, pectin and proteins.

113. Answer (3)

Hint: ER and Golgi apparatus are components of endomembrane system.

Sol.: Materials to be packaged in the form of vesicles from ER fuse with convex face of Golgi apparatus and move towards the concave face.

114. Answer (4)

Hint: Both centriole and flagella in eukaryotes have 9 radial spokes and are made up of microtubules.

Sol.: Eukaryotic flagella shows 9 + 2 arrangement of microtubules while centriole exhibits 9 + 0 arrangement of microtubules.

115. Answer (4)

Hint: In chloroplast, pigments (chlorophyll) are present in thylakoids.

Sol.: Chloroplasts can be lens-shaped, oval, spherical, discoid or even ribbon like organelles. In BGA, chloroplasts are absent.

116. Answer (4)

Hint: Lysosomes are also called suicidal bags.

Sol.: Tonoplast is membrane of vacuole. Vacuole occupy up to 90% cell volume in plants.

117. Answer (2)

Hint: Plasma membrane is selectively permeable in nature.

Sol.: Na^+/K^+ pump is an example of active transport.

118. Answer (3)

Hint: Cytoplasm is the main arena of all cellular activities.

Sol.: Plasmid DNA is used to monitor bacterial transformation.

119. Answer (2)

Hint: Reserve material in prokaryotes are stored in cytoplasm in the form of inclusion bodies.

Sol.: Polysome is not an inclusion body.

120. Answer (1)

Hint: Cristae is infolding of mitochondria, cilia is found in eukaryotic cells and cisternae in ER and Golgi complex.

Sol.: Mesosome increases the surface area for the metabolic activities occurring in prokaryotic cell.

121. Answer (2)

Hint: The elaioplasts store fats and oils.

Sol.: Amyloplasts store starch.

122. Answer (2)

Hint: Centromere appears as a narrow region called primary constriction.

Sol.: Kinetochore is the disc shaped structure present on the sides of centromere.

123. Answer (3)

Hint: Dictyosomes is not a part of chromosome.

Sol.: Telomeres seal the ends of chromosomes and prevent their shortening.

124. Answer (3)

Hint: Middle lamella is made up of calcium and magnesium pectate and hold the adjacent cells together.

Sol.: Plasmodesmata forms the symplastic system between two cells.

125. Answer (3)

Hint: Mitochondria and centrosome are not components of endomembrane system.

Sol.: RER provides the precursors of enzymes for lysosome formation in Golgi complex.

126. Answer (3)

Hint: Cisternae is structural component of ER and Golgi complex.

Sol.: Cristae or inner face of inner membrane of mitochondria is studded with oxysomes.

127. Answer (3)

Hint: This organelle also form spindle fibres that gives rise to spindle apparatus.

Sol.: The centriole is involved in basal body formation in cilia and flagella of eukaryotes.

128. Answer (4)

Hint: Ribosome is non-membrane bound organelle.

Sol.: Lysosome exhibits polymorphism and is single membrane bound.

129. Answer (3)

Hint: Some Golgian vacuole may form lysosome.

Sol.: Lysosome are formed by the process of packaging in Golgi apparatus.

130. Answer (1)

Hint: 60S larger subunit of ribosomes consists of about 40 proteins and three types of rRNAs.

Sol.: 40S smaller subunit consists with 33 protein molecules and a single 18S – rRNA.

131. Answer (2)

Hint: Chromatophores are involved in photosynthesis.

Sol.: Mesosome and mitochondria play important function in respiration.

132. Answer (3)

Hint: Nucleus is known as brain of the cell.

Sol.: Mitochondria is the powerhouse of the cell.

133. Answer (2)

Hint: The organelles whose functions are coordinated form endomembrane system.

Sol.: Centrosome is not included in endomembrane system.

134. Answer (3)

Hint: Gas vacuoles are not surrounded by unit membrane.

Sol.: Gas vacuoles are found in BGA, purple and green photosynthetic bacteria.

135. Answer (4)

Hint: Red blood cells (human) are about 7 μm in diameter.

Sol.: Egg of Ostrich is the largest isolated single cell.

[ZOOLOGY]

136. Answer (4)

Hint: Also called phosphodiester bond

Sol.: Ester bond in DNA, specifically phosphodiester bond, form the covalent backbone connecting adjacent nucleotides. A single phosphate group creates two ester linkages, connecting the 3′ carbon of one deoxyribose sugar to 5′ carbon of the next, providing structural stability to the double helix.

137. Answer (3)

Hint: Inulin contains fruit sugar.

Sol.: Inulin is a natural polymer composed of fructose (fruit sugar), hence called fructans. Its monomeric unit is connected *via* glycosidic bonds. It acts as a reserve carbohydrate in plants, such as Dahlia roots and is commonly classified as a dietary fibre.

138. Answer (3)

Hint: Cellulose is a homopolymer.

Sol.: Cellulose is a polymeric polysaccharide consisting of only one type of monosaccharide *i.e.*, glucose. Starch forms helical secondary structures. Starch can hold I_2 molecules in its helical portion. Cellulose does not contain complex helices and hence cannot hold I_2.

A nucleotide has three chemically distinct moiety. One is a heterocyclic compound, the second is a monosaccharide and the third is phosphoric acid.

The heterocyclic compounds in nucleic acids are the nitrogenous bases and the sugar molecule. The sugar found in polynucleotides is either ribose (a monosaccharide pentose) or 2′ deoxyribose.

139. Answer (2)

Hint: Proteins are heteropolymers.

Sol.: Proteins are not homopolymers. Ribulose bisphosphate Carboxylase-Oxygenase (RuBisCO) is the most abundant protein in the whole of the biosphere. Antibodies are made up of polypeptide chains.

Starch and cellulose are homopolymers.

140. Answer (3)

Hint: Haemoglobin has this kind of protein structure

Sol.: Some proteins are an assembly of more than one polypeptide or subunits. The manner in which these individual folded polypeptides or subunits are arranged with respect to each other (*e.g.*, linear string of spheres, spheres arranged one upon each other in the form of a cube or plate, *etc.*) is the architecture of a protein otherwise called the quaternary structure of a protein.

Quaternary protein is stabilized by four primary forces: Non-covalent interactions, hydrogen bonds, ionic bonds, and disulfide bridges.

141. Answer (4)

Hint: It contains a repeating unit of the same sugar.

Sol.: Chitin is a nitrogen-containing linear polysaccharide and the second most abundant organic substance. It is composed of N-acetylglucosamine.

It forms the exoskeleton of arthropods and cell wall of fungi. Chitin is a homopolymer.

142. Answer (2)

Hint: Malonate *vs* succinate

Sol.: When the inhibitor closely resembles the substrate in its molecular structure and inhibits the activity of the enzyme, it is known as competitive inhibitor. Due to its close structural similarity with the substrate, the inhibitor competes with the substrate for the substrate binding site of the enzyme. Consequently, the substrate cannot bind and as a result, the enzyme activity declines. In competitive inhibition, an inhibitor binds to the enzyme's active site, preventing the substrate from binding. Because the substrate and inhibitor directly compete for the same site, the apparent (K_m) increases, meaning a higher substrate concentration is required to reach half the maximum velocity (V_{max}).

143. Answer (2)

Hint: Both are equal

Sol.: In the formation of a linear polysaccharide chain with 50 glucose units, 49 glycosidic bonds will be formed.

For a chain containing (n) monosaccharide units:-

Number of water molecules removed will be

$\rightarrow n - 1$

Number of glycosidic bonds formed will be

$\rightarrow n - 1$

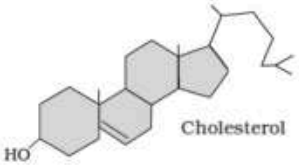
144. Answer (2)

Hint: Found in cell membrane

Sol.: Inorganic catalysts can often withstand extreme conditions.

Physical changes are not catalyzed by enzymes.

Aquaporins are channel-forming membrane proteins that facilitate the rapid transport of water across biological cell membrane.



145. Answer (4)

Hint: Unsaturated fatty acid has comparatively lower melting point.

Sol.: Saturated fatty acid has higher melting point than unsaturated fatty acid.

Palmitic acid is a saturated fatty acid (without double bond) with 16-carbon atoms while, arachidonic acid is a 20-carbon atoms containing unsaturated fatty acid (with double bond).

The absence of double bonds allow their molecular chains to pack closely together in a straight structure.

This tight packing creates stronger intramolecular forces, requiring more heat to melt them.

146. Answer (1)

Hint: In the structure of DNA, [A] pairs with [T]

Sol.: According to Chargaff's rule:

$A = T = 30\% \rightarrow A + T = 60\%$

Remaining $= 40\% \rightarrow G + C$

$\therefore G = C = 20\%$

147. Answer (4)

Hint: Co-factors are crucial for catalytic activity of an enzyme.

Sol.: Catalase – Haem is the prosthetic group.

NADP – Co-enzyme containing vitamin niacin.

Peroxidase – Breaks hydrogen peroxide into water.

Carboxypeptidase – Loses its catalytic activity after removal of its co-factor (Zn^{+2}).

148. Answer (2)

Hint: Oxygen is required for respiration.

Sol.:

Element	**% Weight of**	
	Earth's crust	**Human body**
Hydrogen (H)	0.14	9.5
Carbon (C)	0.03	18.5
Oxygen (O)	46.6	65.0
Nitrogen (N)	very little	3.3
Sulphur (S)	0.03	0.3
Sodium (Na)	2.8	0.2
Calcium (Ca)	3.6	1.5
Magnesium (Mg)	2.1	0.1
Silicon (Si)	27.7	negligible

149. Answer (2)

Hint: Glucose is a hexose sugar.

Sol.: GLUT-4 is the insulin-regulated glucose transporter responsible for facilitating the transport of glucose into the cells.

The one feature common to all those compounds found in the acid-soluble pool is that they have molecular weights ranging from 18 to around 800 daltons approximately.

The acid-insoluble fraction has only four types of organic compounds *i.e.*, proteins, nucleic acids, polysaccharides and lipids. These classes of compounds with the exception of lipids, have molecular weights in the range of ten thousand daltons and above. The chemical compounds which have molecular weights less than one thousand daltons are usually referred to as micromolecules or simply biomolecules.

150. Answer (1)

Hint: Heteropolymers constitute 10-15% of the total cellular mass.

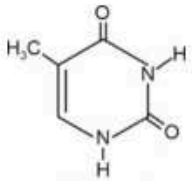
Sol.: Proteins constitute 10-15% of the total cellular mass. Proteins are polymers of amino acids joined together by peptide bonds.

Component	**% of the total cellular mass**
Water	70-90
Proteins	10-15
Carbohydrates	3
Lipids	2
Nucleic acids	5-7
Ions	1

151. Answer (3)

Hint: Uracil is present only in RNA.

Sol.: Thymine is a substituted pyrimidine present in DNA, pairing exclusively with adenine (A) *via* two hydrogen bonds. It provides structural stability to DNA and is replaced by uracil in RNA.



(Thymine)

152. Answer (3)

Hint: Enzymes that catalyse the hydrolysis of peptide bond.

Sol.: Pepsin is a vital gastric enzyme that breaks down proteins into its smaller units. Pepsin belongs to class III (Hydrolases) of enzymes.

Hydrolases are the set of enzymes that catalyse hydrolysis of ester, ether, peptide, glycosidic, C–C, C–halide, P–N bonds, *etc.*

153. Answer (2)

Hint: Specific pocket or cleft in the enzyme's structure.

Sol.: Enzymes are highly specific due to their unique, three-dimensional structure, particularly the shape and chemical environment of their active site. Only specific substrate molecules fit into the active site of the enzyme, similar to lock and key.

154. Answer (3)

Hint: Rubber is a polymeric substance.

Sol.:

Pigments	Carotenoids, Anthocyanins, *etc.*
Alkaloids	Morphine, Codeine, *etc.*
Terpenoides	Monoterpenes, Diterpenes *etc.*
Essential oils	Lemon grass oil, *etc.*
Toxins	Abrin, Ricin
Lectins	Concanavalin A
Drugs	Vinblastin, curcumin, *etc.*
Polymeric substances	Rubber, gums, cellulose

155. Answer (2)

Hint: Molecular weight of lipids is less than 800 Da.

Sol.: Lipids are not considered as polymeric substances as they do not contain long repeating chains of identical/different monomers. While they are large molecules (macromolecules) found in the acid-insoluble fraction, they are formed by smaller, distinct subunits (fatty acids + glycerol) rather than a repeating polymeric chain. Glycosidic bonds (covalent bonds) are present in carbohydrates. Lipids have ester bonds (covalent bonds).

156. Answer (2)

Hint: Insulate the adjacent neurons

Sol.: Reception of stimuli, generation and transmission of nerve impulses are functions of neurons. Neurons, the units of neural system, are excitable cells. The neuroglial cells, which constitute the rest of the neural system, protect and support neurons. Neuroglial cells are non-excitable cells.

157. Answer (1)

Hint: Different types of amino acids are present in proteins.

Sol.: Denaturation of proteins affects secondary and tertiary structures of proteins. Primary structure remains intact during the process of denaturation.

Each protein is a polymer of amino acids.

The amino acids in a protein can be essential or non-essential.

A peptide bond is a covalent linkage that is formed by condensation or dehydration reaction, where the carboxyl group of one amino acid reacts with the amino group of another, releasing a single molecule of water as a byproduct.

158. Answer (1)

Hint: Glycogen is stored in liver.

Sol.: Glycogen is a storage polysaccharide in animals.

Maltose contains two glucose units and has a free aldehyde group hence, it acts as a reducing sugar. As both the aldehyde and ketone groups are involved in glycosidic bond formation, sucrose is a non-reducing sugar.

Cellulose cannot be digested by humans due to lack of cellulase enzyme.

159. Answer (3)

Hint: Also found in the inner lining of fallopian tube

Sol.: Ciliated epithelium is mainly present in the inner surface of hollow organs like bronchioles and fallopian tubes. Their function is to move particles or mucus in a specific direction over the epithelium. Cuboidal epithelium is found in PCT.

160. Answer (2)

Hint: In unicellular organisms, all functions are performed by a single cell.

Sol.: Cells, tissues, organs and organ systems split up the work in a way that exhibits division of labour and contributes to the survival of the body as a whole.

All multicellular animals show division of labour. In unicellular organisms, like *Paramecium*, all functions like digestion, respiration, and reproduction are performed by a single cell and thus, they do not show division of labour.

161. Answer (3)

Hint: More than the number of eyes present in a human

Sol.: Cardiac and skeletal muscle fibres (biceps) are cylindrical in shape and have striated appearance.

- Biceps (skeletal muscles) are multinucleated, unbranched, lack gap junctions and intercalated discs.
- Cardiac muscle fibres are uninucleated, branched and have gap junctions and intercalated discs.

162. Answer (1)

Hint: Bronchioles have ciliated epithelium.

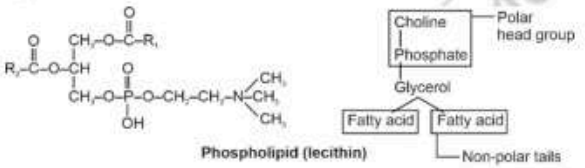
Sol.: The inner surface of the fallopian tube is lined by ciliated columnar epithelium. The function of hair-like cilia is to move the ovum (egg) through the oviduct.

The wall of the fallopian tube is composed of smooth muscle. This muscle performs involuntary peristaltic movements (rhythmic contractions) to assist in transport of the ovum.

163. Answer (3)

Hint: Lecithin is an amphiphilic molecule.

Sol.:



Phospholipid (lecithin)

164. Answer (3)

Hint: Cartilage resists compression.

Sol.: The intercellular matrix of cartilage is solid and pliable and resists compression. Cells of this tissue (chondrocytes) are enclosed in small cavities within the matrix secreted by them. Bones have a hard and non-pliable ground substance rich in calcium salts and collagen fibres which give bone its strength. It is the main tissue that provides structural frame to the body. The bone cells (osteocytes) are present in the spaces called lacunae. Bones interact with skeletal muscles attached to them to bring about movements. The bone marrow in some bones is the site of production of blood cells.

165. Answer (1)

Hint: Major role is protection

Sol.: Compound epithelium is made of more than one layer (multi-layered) of cells and thus, has a limited role in secretion and absorption. They cover the dry surface of the skin, pharynx, inner lining of ducts of salivary gland and of pancreatic ducts.

166. Answer (4)

Hint: Has tall and slender cells

Sol.: The cuboidal epithelium is composed of a single layer of cube-like cells. This is commonly found in ducts of glands, thyroid follicles and tubular parts of nephrons in kidneys and its main functions are secretion and absorption. The columnar epithelium is composed of a single layer of tall and slender cells. They are found in the lining of stomach and intestine and help in secretion and absorption.

167. Answer (2)

Hint: It is a type of dense regular connective tissue.

Sol.: In the dense regular connective tissues, the collagen fibres are present in rows between many parallel bundles of fibres. Tendons, which attach skeletal muscles to bones, and ligaments which attach one bone to another are examples of dense regular connective tissues.

Fibroblasts are present in ligaments.

Bone is the site for production of RBCs.

168. Answer (2)

Hint: Connect cells

Sol.: Adhering junctions: Their primary role is to perform cementing to keep neighbouring cells together. They provide strong mechanical attachment and structural unity to the tissue.

Tight junctions: These are responsible for preventing substances (like ions and molecules) from leaking across the epithelial layer.

Gap junctions: These facilitate communication between cells by connecting the cytoplasm of adjoining cells.

169. Answer (3)

Hint: Cilia move particles or mucus in a specific direction.

Sol.: Compound epithelium is made of more than one layer (multi-layered) of cells and thus, has a limited role in secretion and absorption. Their main function is to provide protection against chemical and mechanical stresses. The columnar epithelium is composed of a single layer of tall and slender cells. Their nuclei are located at the base. The cuboidal epithelium is composed of a single layer of cube-like cells. This is commonly found in ducts of glands and tubular parts of nephrons in kidneys and its main functions are secretion and absorption. The epithelium of proximal convoluted tubule (PCT) of nephron in the kidney has microvilli. If the columnar or cuboidal cells bear cilia on their free surface, they are called ciliated epithelium. Their function is to move particles or mucus in a specific direction over the epithelium.

170. Answer (2)

Hint: Hormones are secreted by endocrine glands.

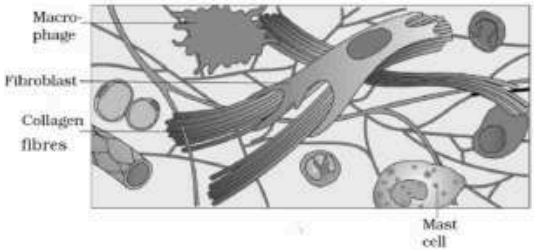
Sol.: Exocrine secretions ⇒ Mucus, saliva, oil, milk and digestive enzymes.

- Moist surface of buccal cavity possesses compound epithelium.
- Endocrine glands secrete hormones.
- Goblet cells of the alimentary canal are unicellular and isolated glandular cells.

171. Answer (2)

Hint: Multiple of 2

Sol.: Erythrocytes (RBCs) and osteocytes (bone cells) are absent in loose connective tissues. Loose connective tissue has cells and fibres loosely arranged in a semi-fluid ground substance. They contain fibroblasts (cells that produce and secrete fibres), macrophages plasma cells and mast cells.



172. Answer (2)

Hint: Proteins efficiently function at the optimum temperature.

Sol.: Mostly enzymes are proteinaceous. Their function is entirely dependent on their specific three-dimensional [tertiary] shape. When exposed to extremely high temperature [195°C], the following occurs:

- Denaturation: High temperature breaks the weak hydrogen bonds and disulfide bridges that maintain the protein's complex folding.
- Once the tertiary structure is disrupted, the enzyme can no longer function properly.
- Proteinaceous enzymes contain peptide bonds. At low temperature, the enzymes get inactivated. Inactivated enzymes can regain their activity.

173. Answer (3)

Hint: Present at locations like alveolar membrane and Bowman's capsule.

Sol.: Squamous epithelium facilitates diffusion and filtration because it consists of a single, ultra-thin layer of flat cells. This minimal thickness creates the shortest possible distance for molecules to travel, allowing for the rapid, passive exchange of gases, nutrients, and waste products.

174. Answer (3)

Hint: The signal is transmitted through synapses between neurons.

Sol.: When a neuron is suitably stimulated, an electrical disturbance is generated which swiftly travels along its plasma membrane. Arrival of the disturbance at the neuron's endings, or output zone, triggers events that may cause stimulation or inhibition of the adjacent neurons and other cells.

175. Answer (4)

Hint: Equal to the number of ears present in a man

Sol.: Mast cells are immune cells found in almost all connective tissues. They produce histamine, heparin and serotonin. They are not found in blood.

176. Answer (1)

Hint: Smooth in appearance

Sol.: The wall of internal organs like stomach, intestine, *etc.*, possess smooth muscle fibres.

The smooth muscle fibres do not show striations and possess cell junctions. Skeletal muscle fibres are multi-nucleated.

177. Answer (3)

Hint: Bone is a specialized connective tissue.

Sol.: Fibres and fibroblasts are compactly packed in the dense connective tissues. Orientation of fibres show a regular or irregular pattern and are called dense regular and dense irregular tissues, respectively. In the dense regular connective tissues, the collagen fibres are present in rows between many parallel bundles of fibres. Tendons, which attach skeletal muscles to bones, and ligaments which attach one bone to another are examples of this tissue. Dense irregular connective tissue has fibroblasts and many fibres (mostly collagen) that are oriented differently.

178. Answer (4)

Hint: Different types of epithelium are present.

Sol.: Epithelial tissue has a free surface, which faces either a body fluid or the outside environment and thus provides a covering or a lining for some part of the body. The cells are compactly packed with little intercellular matrix. There are two types of epithelial tissues, namely, simple and compound epithelium. Simple epithelium is composed of a single layer of cells and functions as a lining for body cavities, ducts, and tubes. The compound epithelium consists of two or more cell layers and has protective function. Squamous, cuboidal and columnar epithelium are present in our body.

179. Answer (1)

Hint: Base + Sugar = Nucleoside

Sol.: Nitrogenous base is a nitrogen-containing ring structure that acts as the alphabet of genetics. There are two main types:

- Purines: Adenine (A) and Guanine (G).
- Pyrimidines: Cytosine (C), Thymine (T) (found in DNA) and Uracil (U) (found in RNA).
- Nucleoside is a structural subunit formed by joining of a nitrogenous base to a five-carbon pentose sugar (Ribose in RNA and Deoxyribose in DNA) *via* a glycosidic bond.
- Nucleotide : It is the foundational building block of DNA and RNA. It is formed when phosphate group is attached to the 5′ carbon of a nucleoside *via* an ester bond.

  Nucleotide = Nucleoside + Phosphate group
- Nucleic acid : A long, continuous polymer formed by linking multiple nucleotide monomers together. The phosphate group of one nucleotide connects to the 3′ carbon sugar of the next, creating a strong sugar-phosphate backbone.

180. Answer (2)

Hint: Supportive cells

Sol.: Neuroglia make up more than one-half the volume of neural tissue in our body.

Chondrocytes are the cells present in cartilage.

Neurons are the structural and functional unit of the nervous system.

Fibroblasts are found in the connective tissues.

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